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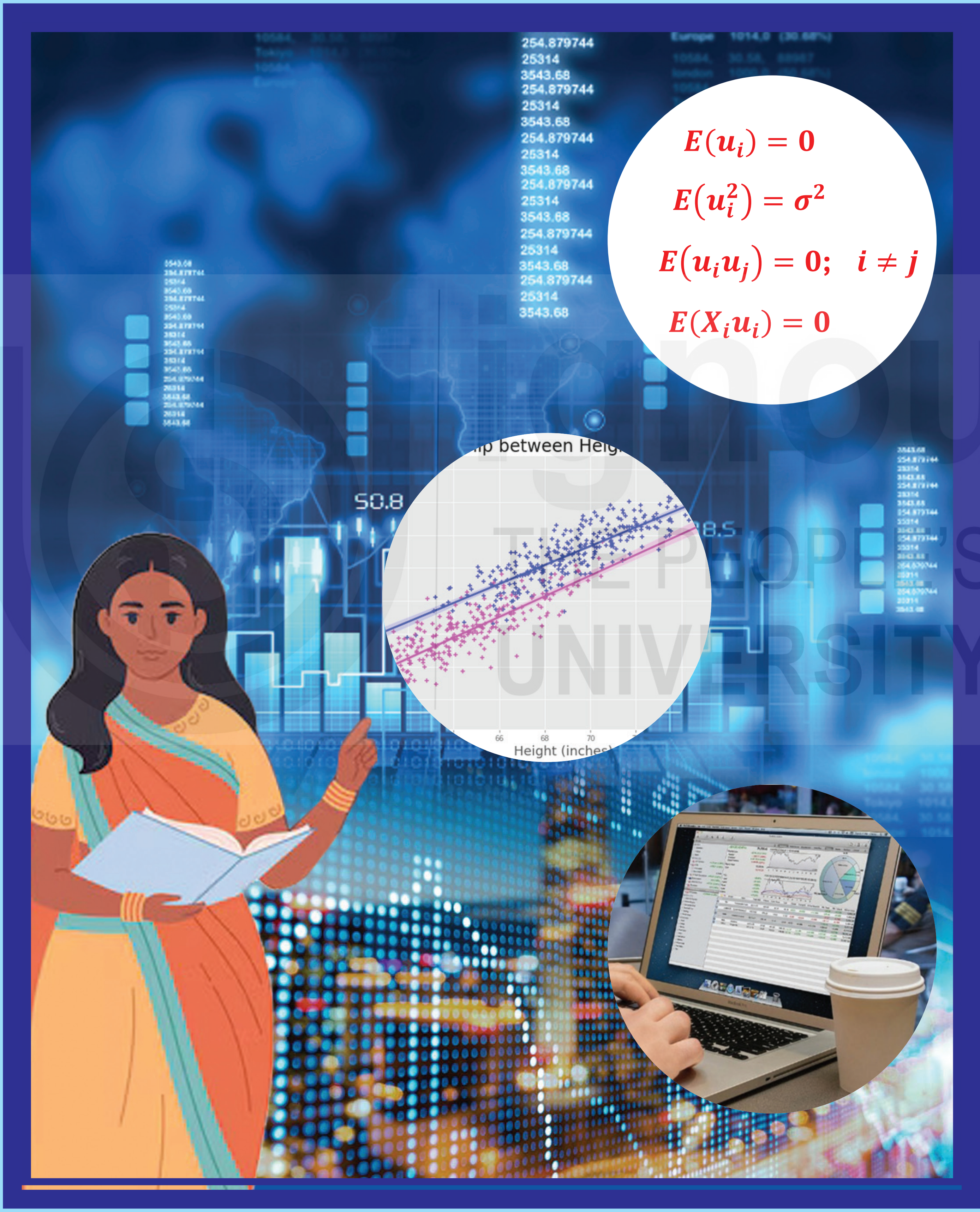
Indira Gandhi National Open University
School of Social Sciences

MECE - 101
INTRODUCTORY ECONOMETRIC METHODS



MECE-101

INTRODUCTORY ECONOMETRIC METHODS

School of Social Sciences
Indira Gandhi National Open University
Maidan Garhi, New Delhi-110068

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CONTENTS

COURSE INTRODUCTION

Econometrics is a powerful tool for understanding economic relationships and for meaningful research in economics. Economists use econometric methods in a variety of fields including macroeconomics, finance, labour economics, health economics, and many more. Econometric methods are used for business decisions, policy formulation, and prediction.

MECE 101, as the title suggests, is introductory in nature. The basic objective of the course is to equip students with basic theory of econometrics and relevant applications of the methods. The topics covered in the present course include various problems faced in estimation of single equation regression models.

After going through the course, a student is expected to

- develop a way of thinking in econometric terms;
- acquire basic econometric skills in terms of model selection, model estimation, and model diagnostics;
- utilize econometric techniques in analysis and interpretation of real-world problems in the field of economics and business; and
- summarise and interpret results of econometric analysis in written, oral and visual forms; and
- develop skills for future research.

The course comprises four blocks.

Block 1, entitled **Introduction**, comprises three Units. As the title suggests, the block provides an overview and basic foundations of econometrics. Unit 1 discusses the basic concepts, issues and ideas in an econometric study. Unit 2 reviews the statistical concepts required in econometric analysis. Unit 3 provides an overview of matrix algebra, which is essential in econometric theory.

Block 2, entitled **Classical Regression Model**, comprises four units. Unit 4 deals with estimation, interpretation and evaluation of two-variable regression models. Unit 5 helps you in identifying problems in a regression model by looking into the residuals. Unit 6 and Unit 7 extend the regression model to multiple explanatory variables cases. By going through these two Units, you will learn the procedure of estimation and evaluation of multiple regression models.

Block3, entitled **Violations of Basic Assumptions**, analyses the properties of the OLS estimators when certain assumptions are not fulfilled. There are six Units in this Block. Unit 8 brings out the specification issues – the possible problems in a regression model – when the basic assumptions of the classical regression model are violated. Unit 9 deals with the identification, consequences and remedial measures if the error terms are correlated. Unit 10 presents the consequences and remedial measures if there is strong correlation between the explanatory variables. Unit 11 delas with the problem of heteroscedasticity where variance of the stochastic error term is not constant across observations. Unit 12 acquaints you with the consequences and remedial measures of errors in dependent and independent variables. Unit 13 discusses the issues when the explanatory variable is correlated with the stochastic error term – the endogeneity problem. In this Unit, the student is introduced to the instrumental variable (IV) method and the two-stage least squares (2SLS) method.

Block 4, entitled **Extensions of Regression Models**, is devoted to two issues – qualitative variables, and simultaneity bias. It comprises three Units. Unit 14 deals with the issues related to the introduction of dummy variables as explanatory variables. Unit 15, on the other hand, deals with issues related to the introduction of dummy variable as dependent variable in regression model. In this Unit, students will get to learn about the logit model and the probit model. Unit 16 introduces the students to possible problems if an equation is estimated by the OLS method, when the equation in fact is a part of an equation system. The issues of endogenous explanatory variables and simultaneity bias are highlighted in this Unit.

After going through MECE 101, students may opt for the course MECE 102: Advanced Econometric Methods. As the name suggests, MECE 102 deals with advanced topics such as time series analysis and panel data models.

UNIT 1 INTRODUCTION TO ECONOMETRICS*

Structure

1.0 OBJECTIVES

After going through this Unit, you would be in a position to

- explain why we should study econometrics;
- appreciate the scope of econometrics;
- point out the important steps in an econometric study;
- distinguish between deterministic relationship and stochastic relationship among variables;
- identify reasons for inclusion of error variable in a regression model;

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- explain the concept of data generation process; and
- transform intrinsically linear models and interpret them.

1.1 INTRODUCTION

Econometrics is a specialised branch of economics which is concerned mostly with empirical verification of economic theory. Let us see how econometrics is different from other branches of economics such as mathematical economics or economic statistics. Mathematical economics usually presents economic theory in mathematical form. It does not bother about empirical measurement of economic theory. Econometrics on the other hand is mainly concerned with empirical verification of economic theory. In doing so, econometrics makes use of the mathematical equations suggested in mathematical economics. Economic statistics deals with collection, processing and presentation of data. Statistics provides many theories on the basis of which inferences can be drawn. For example, the probability rules and sampling distribution are quite important in hypothesis testing. The application of these rules to economic theory through empirical research is a part of econometrics.

In recent years theoretical developments in econometrics have been quite rapid. In fact, it is somewhat difficult to keep track of all these developments. The present unit serves as a foundation for the course. It begins with certain basic concepts and describes how econometric methods are applied to real life situations.

1.2 NATURE OF ECONOMETRICS

Econometrics is a branch of economics which deals with economic measurement. It may be defined as 'the quantitative analysis of actual economic phenomena based on the concurrent development of theory and observation, related by appropriate methods of inference' [Samuelson, Koopmans and Stone, 1954]. We should notice three issues being emphasized in the above statement. First, econometrics deals with quantitative analysis of economic relationships. Second, it is based on economic theory and logic. Third, it requires appropriate methods to draw inferences.

1.2.1 Theoretical Econometrics and Applied Econometrics

Econometrics has both theoretical side as well as applied side. Development of theoretical econometrics has been on two lines: (i) estimation methods, and (ii) test statistics. There are three broad estimation methods: (i) least squares method, (ii) maximum likelihood method, and (iii) method of moments. Under each of these methods, we have several estimation techniques. The least squares method was officially discovered and published by Legendre in 1805 and Carl Friedrich Gauss (1809) made substantial contribution to theoretical advancement of the least squares method. Under least squares method we have many estimation methods such as ordinary least squares, weighted least squares, generalised least squares, indirect least squares, non-linear least squares, etc. We will discuss some of these methods in later Units of this course.

In applied econometrics, we apply these theoretical methods and test statistics to data obtained from various branches of econometrics (such as industrial organisation, labour economics, financial economics, etc.). Very often, the distinction between theoretical econometrics and applied econometrics is blurred – there are several studies which have developed new econometric tools and applied it to empirical data.

Through logic and experience economists have attempted to establish relationship between several variables. Movements in certain economic variables are explained in terms of changes in some other variables. For example, the law of demand states that as price increases the quantity demanded of a commodity decreases. As price of oilseeds increase farmers allocate larger land area for cultivation of oilseeds. You will find numerous examples of such relationships. Some of these are apparent while others are quite complex. As we do not know what exactly is happening we often make conjectures based on logic and our understanding of things.

Due to differences in perception, often different researchers explain the same phenomenon differently. For example, historical data on macroeconomic variables of a country over a period of time are the same. Inferences drawn by researchers on the basis of the same data could be different! While a researcher with classical ideas may come up with the view that government intervention is ineffective, another researcher with Keynesian ideas may suggest the need for government intervention. Thus, the variables included in a model, the nature of relationship among variables, the assumptions made, etc. are quite important.

Some of the economic relationships have been proved through empirical studies and validated under several situations. Some more are put forth as hypothesis and unequivocal conclusions are yet to be drawn. In both the cases, however, logic needs to be supported by empirical observation. An advantage with the subject matter of economics, unlike many other disciplines in social sciences, is that many of the economic variables can be quantified. This has helped in empirical measurement of economic variables and validation of economic theory.

In applied econometrics we deal with empirical studies based on a sample. As you know, it is not possible to carry out a census due to constraints on budget, time and personnel requirements. We consider a single sample and estimate the econometric model (for example, a regression equation). Thus, the population regression function and sample regression function are different. From the estimates obtained from a sample we try to draw inferences about the population. Thus, statistical significance of the estimates is very important in econometric models.

1.2.2 Econometric Models

In economics we often use the term 'model'. It refers to a simplified version of reality. It allows us to understand, analyse and predict economic behavior. An economic model can be for a microeconomic agent such as household or firm. For analysis of production behavior of a firm we assume the existence of production function such as Cobb-Douglas or translog, which determines our supply curve. We assume the structure of the market (perfect competition or oligopoly, for example), which decides the demand curve. By combining the supply curve and the demand curve, we obtain estimates of equilibrium price, output, capacity utilization, etc.

In macroeconomics, economic models represent the behavior of the economy as a whole. Here, we identify relevant macroeconomic variables (such as income, output, expenditure, investment, saving, exports, etc.) and establish a relationship among them. The relationships among these variables may be expressed through diagrams or mathematical equations. There could be macroeconomic models without mathematical expressions, but these may not be precise.

An economic model is based on certain assumptions. These assumptions are required so that minute details are ignored and essential elements are included. Let me illustrate the point through an example. In the case of a firm, it is assumed that there are two factors of production, viz., capital and labour. We combine all types of labour into a homogeneous category – we do not distinguish between a manager and a worker in the field! Similarly, while describing an indifference curve we overlook the type of households – the behavior of a rich household would be different from that of a poor household; or the behavior of a household in a rural area would be different from that of a household in an urban area. We ignore such details because our objective is to analyse the behavior of households to changes in prices and income. If our objective is to identify the changes in consumption pattern across households, we would require a different model and consider such differences.

In the Keynesian model, to take an example from macroeconomics, we consider aggregate variables such as total consumption, total investment, government expenditure, and net exports. We determine equilibrium level of output for the economy as whole. We ignore the behavior of households and firms. Several growth models (such as Harrod-Domar model or Solow model) assume that the economy consists of just one sector – there is an aggregate production function, which gives the relationship between aggregate output (that is, total output) with aggregate inputs (that is, total capital and total labour). It may sound unrealistic, but the objective of these growth models is to analyse the equilibrium conditions for economic growth, saving ratio and population growth. These models ignore the details, but the broad conclusions drawn are helpful in policy formulation. A question such as, 'why growth rate differs across countries?' can be addressed through these growth models.

1.2.3 Important Steps in an Econometric Study

Every econometric study is different in the sense that its objective, methodology, and database are different. The variables included in a study and the methods of estimation of parameters vary across studies. There are, however, certain procedures and sequence which all econometric studies follow. Let us discuss the important steps involved in empirical studies applying econometric methods.

a) The first and foremost issue in econometrics is the statement of hypothesis. Hypothesis is an assertion or claim made about a property or characteristic of a population. Thus, hypothesis is a tentative statement about the research questions of the study. It could be in the form of a simple statement or it could be in the form of relationship among variables. Hypothesis is drawn from economic theory or some logic built by the researcher.

b) The second step is transformation of the above hypothesis into mathematical equation(s). This is our economic model that we want to study. Hypothesis is stated in the form of 'null hypothesis' and 'alternative hypothesis'. Null hypothesis, as you know, present a statement to the effect that there is no relationship between variables.

c) The third step is the collection of relevant data required for the study. Data could be from primary or secondary sources. Usually, data are entered into a worksheet (such as an excel sheet). Data pertains to the variables included in the model.

d) Estimation of parameters of the economic model. This requires selection of appropriate method of estimation. Broadly, there are three methods of estimation: (i) least squares method, (ii) maximum likelihood method, and (iii) method of moments. We will discuss about these methods in subsequent Units. Based on the probability structure of the data available to us, we select the appropriate estimation method.

e) Testing of the hypothesis mentioned in the first step above. Such testing is based on the value of test statistic (such as t-ratio) or the p-ratio. Basically, it amounts to statistical significance of the estimates obtained by us.

f) Interpretation of the economic model. On the basis of inferences drawn through hypothesis testing, we interpret our results.

Check Your Progress 1

1) What is an econometric model?

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2) Distinguish between theoretical econometrics and applied econometrics.

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3) Write down the important steps to be followed in an econometric study.

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1.3 SPECIFICATION OF A REGRESSION MODEL

In economic theory we attempt to establish relationship among variables. Such relationship could be of two types: deterministic and stochastic. Let us focus on the case of two variables: Y and X. As you know from your statistics course, the variables in a regression model are asymmetrical – one variable is dependent (Y) while the other is independent (X).

1.3.1 Deterministic Relationship

The simplest relationship between two variables, Y and X, could perhaps be a linear *deterministic* function given by

$$Y_i = \beta_0 + \beta_1 X_i \qquad (i = 1,2 \ldots\ldots\ldots n) \qquad \ldots (1.1)$$

In the above equation, X is the explanatory variable (or, independent variable) while Y is the explained variable (or, dependent variable). You should note that the subscript i represents the observation number, i ranges from 1 to n, the sample size. Thus, Y_1 is the first observation of the dependent variable, X_1 is the first observation of the independent variable, X_5 is the fifth observation of the independent variable, and so on.

Equation (1.1) implies that Y is completely determined by X and the parameters $\alpha$and β.Suppose we have parameter values β_0 = 3 and β_1 = 0.75, then our linear equation is$Y_i = 3 + 0.75\, X_i$.From this equation we can find out the value of Y for given values of X. For example, when X = 8, we find that Y = 9. Thus, if we have different values of X, then we obtain corresponding Y values on the basis of (1.1).Further, if X_i is the same for two observations, then the value of Y_i will also be identical for both the observations. A plot of Y on X will show no deviation from the straight line with intercept β_0 and slope β_1.

If we look into the deterministic model given by (1.1) we find that it may not be appropriate for describing economic interrelationship between variables. For example, let Y = consumption and X = income of households. Suppose you record your income and consumption for successive months. For the months when your income is the same, does your consumption remain the same? Certainly not, because there are many other factors that decide your monthly consumption expenditure. The point we are trying to make is that economic relationships involve certain randomness. We cannot include all the determinants of monthly consumption expenditure in a regression model. For the sameX_i, we observe different Y_i values.

1.3.2 Stochastic Relationship

The word 'stochastic' means 'random' or 'involving certain probability'. Accordingly, stochastic relationship between X and Y involves certain randomness. In equation form it is given by

$$Y_i = \beta_0 + \beta_1 X_i + u_i \qquad \ldots(1.2)$$

where u_i is the stochastic error term. In equation (1.2) for the same value of X we obtain different values of Y due to the presence of the variable u. In the income consumption relationship, two individuals having the same income can have different expenditure. Stochastic relationship describes real life situation more accurately. Two employees having the same monthly salary in an office can have different monthly expenditure levels. One employee may have a bigger family to support, may have an ailing mother, or an altogether different lifestyle!

Remember that the right-hand side of (1.2) has two parts, viz., i) deterministic part (that is, $\beta_0 + \beta_1 X_i$), and ii) stochastic or randomness part (that is, u_i). Equation (1.2) implies that even if X_i remains the same for two observations, Y_i need not be the same because of different value of u_i. An implication of (1.2) is that if we plot (X,Y) on a graph paper the observations will not remain on a straight line.

It is assumed that the stochastic error term follows certain assumptions:

(i) $E(u_i) = 0$ … (1.3)

While the error is positive for some observations, it is negative for the others. It implies that on an average, the error is zero. In other words, the sum of errors across the population is zero.

(ii) $E(u_i^2) = \sigma^2$ … (1.4)

The variance of the error variable is σ^2. It is constant across the population. For example, the variance of household consumption expenditure is the same for rich households as well as poor households.

(iii) $E(u_i u_j) = 0$ for $i \neq j$ … (1.5)

There is no correlation between the error terms for different observations in a dataset.

(iv) $E(X_i u_i) = 0$ … (1.6)

The explanatory variable is not correlated with the error term. This assumption is required for fulfilment of the properties of the estimators $(\hat{\beta}_0, \hat{\beta}_1)$ on unbiasedness and consistency. You will learn more about the relevance of this assumption in Unit 4.

(v) $u_i \sim N(0, \sigma^2)$ … (1.7)

The error variable u_i follows normal distribution with zero mean and σ standard deviation. For a given value of X, there is a whole range of values of u and a particular value is selected randomly.

The above assumptions are called the classical assumptions. These assumptions indicate the ideal conditions regarding the error variable. In empirical studies, some of the above assumption may not be fulfilled. We will discuss the implications of these assumptions in Unit 4.

1.3.3 Inclusion of Error Variable

In the previous sub-section we discussed about the statistical properties of the error variable (u_i). Now, let us find out the logic behind inclusion of the error variable in the regression model. There are at least three reasons for inclusion of the stochastic error term in a regression model.

First, in real life the relationship among variables is complex. Regression model is a simplification of such relationships. It is not always possible to include all relevant variables in the functional relation. For example, we assume that price is the sole determinant of demand for a product. In fact, several variables related to demand, such as individual tastes, population, income, etc. do influence the price of a product, which are omitted from the function. Some of these variables have positive effects while others have negative effects. The net effect of these omitted variables in the equation is represented by the error term. If the effect of these omitted variables is small, it is reasonable to assume that the error term is random.

Second, we do not know the exact functional form. Suppose we hypothesize a linear relationship among the variables, while it is a quadratic one. There is no means of knowing what the true relationship is. We just take one functional form on basis of our logic and reasoning. Certain error creeps into the model, because of the wrong specification of the functional form. Presence of an error term absorbs all such misspecification issues.

Third, another source of error is the sampling error. For instance, consider equation (1.2) as a household-consumption function, where Y is consumption and X is income. Even if equation (1.2) is a correct relationship, the sample we randomly choose to examine may turn out to be predominantly poor families. Thus, estimates of β_0 and β_1 from this sample group may not be as good as the estimates obtained from a representative sample.

1.4 DATA GENERATION PROCESS

By now you must be clear that econometric studies require specification of the model in quantitative terms, involving variables and parameters. In this context, selection of relevant variables and specification of the functional form are quite important.

Let us examine the consumption-income model to illustrate the above point. We observed that monthly consumption expenditure of two households having the same monthly income (say, Rs. 50000) are different – while one household consumes Rs. 35000, the other household consumes Rs. 48000.After considering its income level and other determinants of household expenditure, a household decides on its consumption level. What are these other determinants of household expenditure? These could be family size, health status, culture, tastes, etc. all these determinants collectively determine the consumption expenditure of a household. Do we know all the relevant variables? Is it possible to include all these variables in our regression equation?

While specifying a consumption function, it is not possible to include all the determinations of consumption expenditure. Further, as pointed out in Sub-Section 1.2.2, an econometric model is a simplification of reality. Thus, in an econometric model we include the important variables only.

From our sample survey (or, from the secondary data sources) we obtain data. A major issue before us is our ignorance of the data generation process (DGP). Such data can be seen as a joint probability distribution with certain probability distribution. As mentioned above, we do not know all the factors that influence the consumption decision of a household. We make some guesswork, based on our experience and logic. Accordingly, we assume a probability structure for the data.

Data generation process (DGP) reflects the reality that is taking place. As DGP is not known to us, we try to uncover the secrets through our econometric models. There are two approaches to econometric modelling: (i) general-to-specific, and (ii) specific-to-general. In the first approach we consider a general model by including many variables. Subsequently, we exclude the irrelevant variables. In the second approach, we begin with a simple model and keep on adding variables based on our logic. Although, there is no consensus on the approach to be followed, researchers prefer the general to specific approach.

There are four steps in the process of building an econometric model: (i) we guess the data generation process, (ii) we assume the associated probability theory, (iii) we use that theory for empirical evidence, and (iv) we revise the model if the results do not match our data. We continue with the above four steps in an iterative manner till we get satisfactory results.

We make several trial and error on the functional form. If the data fails to fit our model, it implies we do not understand the data generation process. We try with another model. If the data fits our model, we still do not understand the data generation process! It just implies that our model was good in this case.

You should recall how a hypothesis is formulated. It is a statement based on theory or logic. If we could reject a null hypothesis, we confirm that the dataset does not hold true (recall Karl Popper's theory of falsification discussed in MEC 109).

1.5 FUNCTIONAL FORMS

Usually, we consider a linear regression equation in our econometric model, as it is easier to estimate and interpret such models. Linear regression models are linear in parameters and linear in variables. Linear regression models, however, may not be appropriate in all cases; there could be situations where the regression model is non-linear.

1.5.1 Classical Regression Model

The two-variable linear regression model is the simplest functional form that we consider. It is given by

$$Y_i = \beta_0 + \beta_1 X_i + u_i \qquad \ldots(1.2)$$

where Y is the explained variable (also called dependent variable)

X is the explanatory variable (also called independent variable)

u is the error variable (also called white noise in time series analysis)

When we include more than one explanatory variable in the regression model, we call it the multiple regression model. In notations, a multiple regression model is given by

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + u \qquad \ldots(1.8)$$

In equation (1.8) there is one explained variable (Y) and k explanatory variables ($X_1, X_2 \ldots, X_k$). We have omitted the sub-scripts for simplicity.

When the error variable u fulfils all the classical assumptions specified at equations (1.3) to (1.7), we call the regression model a 'classical regression model'. Ordinary Least Squares (OLS) method is frequently used to estimate parameters of the classical regression models.

1.5.2 Intrinsically Linear Cases

Certain non-linear equations can be transformed into linear equations by taking logarithms. Two intrinsically linear models are used frequently: (i) semi-log model, and (ii) log-linear models.

(a) Semi-Log Models

Let us begin with a functional form as follows:

$$Y_t = e^{\beta_0 + \beta_1 X_t + u_t} \qquad \ldots(1.9)$$

This regression model, in its present form, is non-linear. Therefore, it cannot be estimated by OLS method. However, if we take natural logs of both the sides, we obtain

$$\ln Y_t = \beta_0 + \beta_1 t + u_t \qquad \ldots(1.10)$$

It transforms into a semi-log equation. It is called a semi-log model as one of the variables is in log form.

If we take $\ln Y_t = Y_t'$, then equation (1.10) can be written as

$$Y_t' = \beta_1 + \beta_2 t + u_t \qquad \ldots(1.11)$$

Estimation of equation (1.11) is simple. The equation is linear in parameters and in variables. Thus, we can apply OLS method to estimate the parameters. The implication of the regression model (1.11), however, is much different from the regression model (1.2).

If we take the differentiation of the regression model $Y_t = \beta_0 + \beta_1 X_t + u_t$, we obtain

$$\frac{dY}{dt} = \beta_1 \qquad \text{… (1.12)}$$

Equation (1.12) shows that the slope of the regression equation is constant. An implication of the above is that the absolute change in the dependent variable for unit increase in the independent variable is constant throughout the sample. If there is an increase in *X* by one unit, *Y* increases by β_1 unit.

Now, let us consider the regression model $\ln Y_t = \beta_0 + \beta_1 t + u_t$. If we take differentiation of equation (1.11) we find that

$$\frac{d\ln Y_t}{dt} = \beta_1$$

which means

$$\frac{1}{Y_t}\frac{dY_t}{dt} = \beta_1 \qquad \text{… (1.13)}$$

An implication of equation (1.13) is that the slope of the regression model is variable. Thus, its interpretation is different from that of the regression model $Y_t = \beta_0 + \beta_1 X_t + u_t$.

For equation (1.13), we interpret the slope coefficient (β_1) as follows: For every unit increase in *X*, there is β_1 *per cent* increase *Y*. Thus, for a semi-log model the change in the dependent variable in terms of percentages. The semi-log model is useful is estimating growth rates.

(b) Log-Linear Models

Let us consider the following non-linear regression model

$$Y = \beta_0 X^{\beta_1} \qquad \text{… (1.14)}$$

This model will be intrinsically linear if it can be transformed into

$$Y' = \beta_0 + \beta_1 X' + u \qquad \text{… (1.15)}$$

Using the logarithm of each of the variable in equation (1.13), we get the following transformed equation:

$$\ln Y_i = \beta_0 + \beta_1 \ln X_i + u_i \qquad \text{... (1.16)}$$

The regression model given at (1.16) is called a log-linear model (because it is linear in logs of the variables) or double-log model (because both variables are in log form).

Let us take differentiation of equation (1.16) with respect to $\ln X_i$

$$\frac{d(\ln Y_i)}{d(\ln X_i)} = \beta_1 \qquad \text{... (1.17)}$$

A closer look at equation (1.17) shows that the slope parameter of (1.14) represents the elasticity between *Y* and *X*. This attractive feature of the log-linear model has made it popular in applied work. The slope coefficient β_1 measures the elasticity of *Y* with respect to *X*, that is, the percentage change in *Y* for one per cent change in X. Thus, if Y represents the quantity of a commodity demanded and X its unit price, then β_1 measures the price elasticity of demand.

The log-linear regression model has certain advantages: (i) the parameters are invariant to change of scale, since they measure percentage changes, (ii) the model gives elasticity figures directly, and (iii) the model moderates the problem of heteroscedasticity to some extent (see Unit 11 for the problem of heteroscedasticity).

1.6 SOFTWARE PACKAGES FOR ECONOMETRIC ANALYSIS

As econometric analysis deals with empirical data it involves cumbersome computations. We can think of estimating the parameters in some of the simpler estimation methods (such as OLS) involving a small dataset. In other cases, however, it is quite difficult and time consuming.

In recent years, easy access to computers and the availability of econometric software packages has made the job simpler. We find that the most difficult computations can be done by computers in a few seconds. There are several software packages available for econometric applications. Most of these software packages include the important econometric applications. Further, since these software packages apply similar methods, the results are almost the same. Some of these software are freely available while others need to be bought from the market. An open-source software package (i.e., freely downloadable from the Internet)is R. You can modify this software according to your need share with others without any licensing violation. Thus, a lot of new features (i.e., new methods and test statistics) are being added to this software. A major limitation of R is that you need to have some basic knowledge of computer programming.

Software such as STATA, E-Views and SPSS are more user-friendly in the sense that you can apply various econometric methods by the click of a mouse – you do not have to write programs. These software, however, are licensed and need to be purchased. There are certain freeware packages such as Gretl, EasyReg, Matrixer, etc. which can be freely downloaded from the Internet and can be used for econometric analysis.

Check Your Progress 2

1) Explain the need for inclusion of an error variable in a regression model.

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2) Explain the concept of data generation process. Is it the same as the estimated regression model?

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3) How do you interpret the slope parameter of the following regression models?

(i) $Y_i = \beta_0 + \beta_1 X_i$

(ii) $\ln Y_t = \beta_0 + \beta_1 t + u_t$

(iii) $\ln Y_i = \beta_0 + \beta_1 \ln X_i + u_i$

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1.7 LET US SUM UP

In this Unit we discussed the importance and scope of econometrics. Also, we distinguished between theoretical econometrics and applied econometrics. In econometric models we follow certain steps: specification of the model in quantitative terms, collection of data, estimation of the model, testing of hypothesis and interpretation of results, and drawing inferences.

Econometric models are stochastic in nature; we add a stochastic error variable to the regression model. The stochastic error is a random variable. It includes the omitted explanatory variables, error due to wrong specification of the functional form, and selection of wrong sample.

Data that we consider for estimation of parameters take occur through a data generation process (DGP). The DGP is not known to us; we try to conjecture it through some econometric model. We consider several alternative models and opt for the best out of these alternative econometric models.

1.8 KEY WORDS

Alternative Hypothesis : In hypothesis testing, alternative hypothesis states a condition that is opposite to the null hypothesis. It is expressed as $H_1: \beta_2 \neq 0$, i.e., the slope coefficient is different from zero. It could be positive or negative.

Classical Linear Regression Model : It refers to a linear regression model that establishes a linear relationship between the variables, based on certain assumptions regarding the error variable.

Confidence Interval : It is the range of values that determines the probability that the value of the parameter lies within the interval.

Deterministic : It represents the systematic component of the regression equation. It is the expected value of the

Component		dependent variable for given values of the explanatory variable.
Econometric Model	:	These are statistical models specifying relationship between relationships between various economic quantities.
Estimation of Parameters	:	This process deals with estimating the values of parameters based on empirical data that has a random component.
Null Hypothesis	:	It is the hypothesis that there is no significant difference between specified population, the observed difference is mainly due to sampling or experimental error.
Parameter	:	A quantity or statistical measure for a given population that is fixed. The mean and the variance of a population are population parameters.
Population Regression Function (PRF)	:	A population regression function hypothesizes a theoretical relationship between a dependent variable and a set of independent or explanatory variables. It is a linear function. The function defines how conditional expectation of a variable Y responds to the changes in independent variable X.
Statistical Inference	:	It refers to the process of deducing properties of underlying probability distribution of the parameters by analysing data.
Stochastic Error	:	The error variable represents the influence of those variables that are not included in the regression model. It is evident that even if we try to include all the factors that influence the dependent variable, there exists some intrinsic randomness between the two variables.

1.9 SOME USEFUL BOOKS

Maddala, G. S., and Lahiri, K. (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley & Sons.

Verbeek, M. (2017), *A Guide to Modern Econometrics.* Fifth Edition, John Wiley & Sons.

Wooldridge, Jeffrey M. (2022), *Introductory Econometrics: A Modern Approach*, Cengage Learning.

1.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Econometric model is a simplification of reality. It retains the major features while excludes the finer details. It is expressed in mathematical form. It can be estimated by using real life data.
2) Refer to Sub-Section 1.2.1 and answer.
3) Refer to Sub-Section 1.2.3 and answer.

Check Your Progress 2

1) The error variable signifies the stochastic nature of relationship. The error term includes the omitted explanatory variables, error due to wrong specification of the functional form, and error due to selection of wrong sample.
2) Data generation process reflects the reality. It indicates the process through which values of different variables pertaining to an observation occur. The DGP is not known to us. We try to approximate it with certain econometric model.
3) (i) A unit change in X results in β_1 units of change in Y.

   (ii) A unit change in results in β_1 per cent of change in Y.

   (iii) A unit per cent change in X results in β_1 per cent of change in Y.

UNIT 2 REVIEW OF STATISTICAL CONCEPTS*

Structure

2.0 OBJECTIVES

After going through this unit, you will be able to

- differentiate between a parameter and a statistic;
- distinguish between a 'point estimate' and an 'interval estimate';
- discuss the properties of a 'good estimator';

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- write a note on the asymptotic properties of an estimator;
- define the concepts of 'weak and strong law of large numbers';
- state the 'Central Limit Theorem' illustrating the same with an example;
- explain the different terminologies used in the context of the testing of a 'statistical hypothesis';
- apply the least squares method to estimate the parameters of a linear regression model;
- elucidate the method of 'maximum likelihood estimation (MLE)';
- derive the 'maximum likelihood estimates' of the parameters of standard distributions like Binomial, Poisson and Normal;
- list the properties of MLE; and
- describe the 'method of moments' estimation with illustrations.

2.1 INTRODUCTION

Estimation and hypothesis testing are the two parts of classical statistical inference that is employed all through in econometrics. The process of estimating a population parameter by a random sample drawn appropriately from a population is referred to as 'estimation'. The process of assessing whether to accept or reject a hypothesis, making an assumption about the value of a population parameter based on sample information, is done in 'hypothesis testing'. Inferences (or conclusions) about the relationship between a dependent variable and independent variable, in probabilistic terms, is done in hypothesis testing of regression models (or equations) fitted from sample data.

The aggregate of objects under consideration in any statistical research is referred to as the 'population'. In sampling theory, population refers to the 'target group' from which the samples are drawn. A sample is a finite portion of the population selected with the objective of exploring its properties. The sample size refers to the number of units in the sample. The statistical constants of the population like mean (μ), variance (σ^2), skewness (β_1), Kurtosis (β_2), moments (μ) correlation coefficient (ρ) etc are known as 'parameters'. We compute corresponding statistical constants for the sample drawn from the population. Such statistical constants of the sample such as mean ($\bar{x}$), variance (s^2), skewness (b_1), Kurtosis (b_2), moments (m_r), and correlation coefficient (r) are referred to as 'statistics'.

2.2 STATISTICAL INFERENCE

The purpose of sampling is to investigate a characteristic of the population based on sample observation. A carefully selected sample is supposed to reveal their characteristics. Thus, in statistical inference, we deduce information about the population (more precisely about the population parameters) based on an analysis of the sample. This process is known as 'statistical inference'. To begin with, we may have no information about the population in terms of their parametric values. Therefore, estimation is required. Secondly, some data or hypothetical values for the parameter may be available, and it may be required to test whether a hypothesis formulated is tenable. This brings us to the need for 'hypothesis testing'. The estimation may be of two types, viz., (i) point estimation, and (ii) interval estimation.

2.2.1 Point Estimation

A point estimate of a parameter is a specific value of a 'statistic'. The term 'statistic' here refers to the aggregate value of sample observations (e.g. mean, standard deviation). Thus, the term 'statistic', more generally refers to a function of sample observations. The value obtained by computing the statistic (e.g. $\bar{x} = \frac{x1+x2+\cdots xn}{n}$), for a particular sample is called the 'estimate'. Thus, if (x_1, $x_2 \ldots x_n$) is a random sample from the population and the statistic is more generally denoted by 'T', then

$$T = f(x_1, x_2 \ldots x_n) \qquad \ldots (2.1)$$

is the computed value of T using the sample data $x_1, x_2 \ldots x_n$. In other words, T is the 'estimator' and the value taken by it for the sample observations, say t, is an 'estimate'. An estimator is thus a general rule or a formula for estimating an unknown population parameter. Thus estimator (i.e., the formula) will be the same for multiple samples. The estimate (i.e., the value obtained by applying the formula) will vary from sample to sample. Any particular estimate obtained from a sample is a 'point estimate'. The distribution of many such point estimates, obtained from different samples, is called the 'sampling distribution' of the 'statistic'.

2.2.2 Interval Estimation

Let $Ø$ be a population parameter and let T_1 and T_2 be two functions based on sample observations such that '$P(T_1 \leq Ø \leq T_2) = (1 - \alpha)$'. Here, α is the level of significance (usually taken as a low value of 1 or 5 percent). Now, for the parameter $Ø$, with the confidence coefficient as '$1 - \alpha$', the interval (T_1, T_2) is termed as an 'interval estimate' or a confidence interval. The statistics T_1 and T_2 are called the lower and upper confidence limits for the parameter $Ø$. More specifically, to illustrate, if we draw a random sample (X) from a population whose parameter values (mean, variance) are not known, and the sample observations are 2, 3, 5, 6 & 9, then the sample mean which is $\bar{X} = 5$ represents the point estimate of the unknown population mean. However, if we say the true population mean lies between ($\bar{X} - 0.5$) and ($\bar{X} + 0.5$), i.e., between 4.5 and 5.5, then the range of values from 4.5 to 5.5 becomes an 'interval estimate' of the 'population mean'.

2.2.3 Properties of a Good Estimator

There are four criteria for a good estimator. These are: (i) unbiasedness, ii) consistency, iii) efficiency and iv) sufficiency. Let us formally define each one of this now.

Unbiasedness: A statistic 'T' is said to be an unbiased estimator of a parameter $\emptyset$, if the expected value of T is $\emptyset$, *i.e.*,

$$E(T) = \emptyset \qquad \ldots (2.2)$$

As an illustration of this property, we can see how the sample mean $\bar{x}$ is an unbasied estimator of the population mean μ. We have: sample mean: $\bar{x} = \frac{x1+x2+\cdots.xn}{n}$ and the population mean $(\mu) = \frac{X1+X2+\cdots.+Xn}{N}$. We need to verify that $E(\bar{x}) = \mu$. Each of the sample member x_1, $x_2 \ldots x_n$ is a random variable because their values in any particular sample depends on its probability of being included in the sample. For instance, in SRSWR any of the population members X_1, $X_2 \ldots X_n$ may appear at the 'i^{th}' drawing, *i.e.*, x is the random variable with the probability distribution as follows:

X_i	X_1	X_2	....	X_n
Probability	$\frac{1}{N}$	$\frac{1}{N}$	....	$\frac{1}{N}$

$$E(x_i) = X_1.\frac{1}{N} + X_2.\frac{1}{N} + \ldots\ldots + X_n.\frac{1}{N}$$

$$= \frac{1}{N}(X_1 + X_2 + \ldots. + .X_n)$$

$$= \mu = \text{population mean}$$

$$E(\bar{x}) = E\left[\frac{1}{n}(X_1 + X_2 + \cdots + X_n)\right]$$

$$= \frac{1}{n}[E(X_1) + E(X_2) + \ldots. E(X_n)]$$

$$= \frac{1}{n}[\mu + \mu + \ldots + \mu] = \frac{1}{n}n\mu = \mu$$

Minimum Variance Unbiased Estimator (MVUE): Since the sample drawn can be many, and each sample has its own variance, this needs us to consider the term 'minimum variance unbiased estimator'. The minimum variance unbiased estimator will have the smallest variance in the class of all unbiased estimators. In other words, if T_m is the minimum variance unbiased estimator of any population parameter $\emptyset$, then $E(T_m) = \emptyset$ and $Var(T_m) < Var(T)$, for any T with $E(T) = \emptyset$.

Consistency: Let $X_1, X_2, \ldots, X_n$ be the random variables drawn from a population with parameter $\emptyset$. Let 'T' be a statistic computed as an estimate of the population parameter $\emptyset$. The statistic 'T' is said to be a consistent estimator of the parameter $\emptyset$, if:

$$\lim_{n \to \infty} P(T \longrightarrow \emptyset) = 1 \qquad \ldots (2.3)$$

Note that for any distribution, sample mean ($\overline{X}$), is a consistent estimator of the population mean (μ) and sample variance S^2 is a consistent estimator of the population variance σ^2.

Efficiency: The most efficient estimator is one which has minimum variance among all consistent estimators. If T_1 and T_2 are consistent estimators of a parameter $\emptyset$, such that

$$\text{Var}(T_1) < \text{Var}(T_2), \forall n$$

then T_1 is said to be more efficient than T_2. If T_1 is the most efficient estimator with variance V_1 and T_2 is the other estimator with variance V_2 for a parameter $\emptyset$, then the efficiency of the estimator is defined as:

$$\text{Efficiency} = \frac{V_1}{V_2}$$

Since $V_1 \leq V_2$, the efficiency of any estimator cannot exceed unity.

Sufficiency: Let $X_1, X_2 \ldots\ldots X_n$ be the random variables of size n drawn from a population with parameter $\emptyset$. Let 'T' be a statistic computed to estimate the parameter $\emptyset$. If 'T' contains complete information in the sample regarding the parameter $\emptyset$, then it is called a sufficient estimator. Now, if a statistic 'T' is computed on the basis of a random sample of size n, from a population with probability density function (p.d.f) $f(X, \emptyset)$, then T is a sufficient estimator of $\emptyset$ if the conditional distribution of $X_1, X_2 \ldots Xn$ for a given value of T is independent of $\emptyset$. This means, the conditional probability P [$X_1 \cap X_2 \cap \ldots \cap X_n \mid T = t$] does not depend on (or contain) $\emptyset$.

Mean Squared Error: When comparing two estimators, a need to make a choice between the two may arise. Suppose we have two estimators, one of which has lower bias but more variance than the other. By this, we mean:

$(bias\ T1) > (bias\ T2)$ but $Var(T1) < Var(T2)$.

Now, the mean-squared error for T_1 is defined as:

$$MSE(T1) = E[T1 - \emptyset]^2 = Var(T1) + (biasT1)^2$$

Now, if MSE (T_1) < MSE (T_2), we say that T_1 has a lower mean-squared error and accept it as an estimator of $\emptyset$ based on the mean-squared error property.

2.3 ASYMPTOTIC PROPERTIES OF AN ESTIMATOR

The asymptotic properties relate to an estimator's distribution as the sample size grows larger and approaches infinity. The following are the two asymptotic properties of an estimator.

Asymptotic Unbiasedness: An estimator T is an asymptotically unbiased estimator of $\emptyset$ if: $\lim_{n\to\infty} E(T) = \emptyset$. It means, as the sample size approaches infinity, the estimator T, which was previously biased, becomes unbiased.

Consistency: The asymptotic property of consistency relates to the behaviour of an estimator's bias and variance as the sample size approaches infinity. If increasing the sample size reduces the estimator's bias and variance, with the trend continuing till both the bias and variance reaches zero as n $\to\infty$, the estimator is said to be consistent. Thus, for T to be a consistent estimator, it must satisfy the following two conditions: (i) $\lim_{n\to\infty} [E(T) - \emptyset] = 0$, and (ii) $\lim_{n\to\infty} Var(T) = 0$.

2.3.1 Law of Large Numbers

Let $X_1, X_2, \ldots, X_n$ be a sequence of i.i.d. (independent and identically distributed) random variables each with mean μ and standard deviation σ. Let the sample mean be given by: $A_n = \frac{X_1 + X_2 + \cdots + X_n}{n}$. Now, the weak law of large numbers states that for all α > 0: $\lim_{n\to\infty} P(|A_n - \mu| > \alpha) = 0$ [or $\lim_{n\to\infty} P(|A_n - \mu| < \alpha) = 1$]. The strong law of large numbers, on the other hand, requires that $\lim_{n\to\infty} P(A_n = \mu) = 1$ [or $\lim_{n\to\infty} P(|A_n - \mu| = 0) = 1$].

The concept of sampling distribution of a statistic can now be revisited as follows. If $x_1, x_2, \ldots, x_n$ are random variables of same sample size n drawn from a population with their respective means as $\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n$, the sampling distribution of these means is the 'probability distribution of these sample means'. If we further assume that the samples are drawn from a normal population, with the n samples drawn are 'independent and identically distributed' (*i.i.d.*), it leads us to a result called the 'central limit theorem' discussed next.

2.3.2 Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be n 'independent and identically distributed' (*i.i.d.*) random variables, each of size n, and drawn from a population having mean μ and variance σ^2, then as the sample size n becomes large, the distribution of the sample mean $\bar{X}$ approaches normal distribution with mean μ and variance $\frac{\sigma^2}{n}$. In other words, we have: $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ as $n \longrightarrow \infty$. You are familiar with the concept of standardised normal variable. It is obtained by subtracting the mean from the variable and dividing it by the standard deviation such that

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \qquad \ldots (2.4)$$

The variable Z as defined in equation (2.4) approaches to the stranded normal distribution, i.e., a normal variate with zero mean and unit variance, as n becomes large. Let us consider an illustration now.

We wish to test whether a coin tossed 30 times, with 'Heads' obtained in 13 tosses is unbiased. Here, n = 30, and $\bar{x} = \frac{13}{30} = 0.43$. If the coin is unbiased, then b

= 0.5. Since it is a case of coin tosses with two outcomes (H and T), the binomial probability distribution is applicable. As you know, in the case of binomial distribution, the variance σ^2 is *npq*. Thus, $\sigma = \sqrt{npq}$. Hence, the term with $\sqrt{n}$ in the denominator cancels out. The central limit theorem (CLT) as per equation (2.4) tells us that

$$Z = \frac{\bar{X} - 0.5}{\sqrt{pq}} \sim N(0, 1)$$

$$Z = \frac{0.43 - 0.5}{\sqrt{0.5(1-0.5)}} = \frac{-0.07}{0.5} = -0.14$$

The critical value of Z at 5 per cent level of significance is 1.96. The Calculated value of Z (absolute) is less the critical Z value. We therefore accept the null hypothesis that the coin is not biased. We conclude that the data from the trials does not suggest that the coin is biased.

Check Your Progress 1

1) Differentiate between a parameter and a statistic.

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2) Distinguish between a 'point estimate' and an 'interval estimate'.

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3) What is meant by the term 'sampling distribution of a statistic'?

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4) State the properties of a 'good estimator'.

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5) What are the two conditions that a consistent estimator should satisfy?

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6) Differentiate between the weak and strong versions of the law of large numbers.

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2.4 HYPOTHESIS TESTING

A statistical hypothesis is a statement about a population or equivalent about a population parameter. Such a statement made may or may not be true. Hence, we wish to test the hypothesis on the basis of the evidence from a random sample. If the hypothesis completely specifies the population, then it is known as a 'simple hypothesis'; else, it is known as a 'composite hypothesis'. For instance, in a sampling from a normal population $N(\mu,\sigma^2)$, the hypothesis H: $\mu = \mu_0$, $\sigma^2 = \sigma_0^2$ is a simple hypothesis. It is complete because it specifies a value for both the parameters of the distribution. On the other hand, each of the following hypothesis is composite hypothesis.

i) $\mu=\mu_0$ (nothing is specified about σ^2)

ii) $\sigma^2 = \sigma_0^2$ (nothing is specified about μ)

iii) $\mu < \mu_{0,}\ \sigma^2 = \sigma_0^2$ (μ is not exactly specified).

2.4.1 Null Hypothesis and Alternative Hypothesis

The null hypothesis is an assertion whose validity is to be tested for possible rejection based on sample observations. It is denoted by $H_{0.}$ It is tested against other options specified in the 'alternative hypothesis'. An alternative hypothesis is indicated by H_1. The alternative hypothesis is not tested, but its acceptance or rejection is determined by the null hypothesis's rejection or acceptance. In a strict sense, we actually either 'reject' or 'do not reject' a null hypothesis. Though the latter (i.e., do not reject) is commonly referred to as 'acceptance', it is actually the case of 'not rejecting the null hypothesis'. Therefore, you should always remember that a hypothesis is never accepted; it is either rejected or not rejected.

The null hypothesis that the population mean μ takes a specific value μ_0 is expressed as H_0: $\mu= \mu_{0.}$ Now, the alternative hypothesis may be expressed as: H_1: $\mu \neq\mu_0$ or H_1: $\mu > \mu_0$ or H_1: $\mu < \mu_{0.}$ This brings us to the concepts of 'one-tailed test' and 'two-tailed test'. A null hypothesis tested against the alternative hypothesis like $\mu < \mu_0$ or $\mu > \mu_0$ is called a one-tailed test. More specifically, if the alternative hypothesis is H_1: $\mu > \mu_o$ it is called a right tailed test. Likewise, if the alternative hypothesis is H_1: $\mu < \mu_o$ it is called a left-tailed. Both these are cases of 'one-tailed test'. A null hypothesis where the alternative hypothesis is H_1: $\mu\neq \mu_o$ is known as 'two-tailed test'.

2.4.2 Type-I Error and Type-II Error

The error committed in rejecting a null hypothesis by the test, when it is actually true, is termed as Type-I error. The error committed in accepting a null hypothesis by the test, when it is actually false, is termed as Type-II error. The correct decision by the test procedure followed, and the likely error, is depicted in Table 2.1.

Table 2.1: Error in Hypothesis Testing

	Ho is True	Ho is False
Ho is accepted	✓	Error of Type-II
Ho is rejected	Error of Type-I	✓

2.4.3 Level of Significance

The level of significance refers to the maximum permitted level of committing Type-I error. It is a level of risk that we are willing to take. It is denoted by $\propto$ and indicates the probability of rejecting the null hypothesis [i.e., P (rejecting Ho when Ho is true) = $\propto$]. The most commonly used levels of significance in practice are 5 per cent and 1 per cent. If we use a 5 per cent level of significance, this means that we are likely to reject a 'right Ho' in 5 out of 100 samples. To put it another way, this means we have a 95 per cent confidence level of not committing the Type-I error. Thus, at $\propto$ level of significance, the degree of confidence in our decision is $(1-\propto)$ which is also called the 'confidence coefficient'. The *p*-value is thus defined as the lowest significance level at which a null hypothesis can be rejected.

2.4.4 Critical Region

The critical region indicates the probability that the Type-I error committed in our statistical inference does not exceed the level of significance. Let us consider the drawing of numerous samples of the same size from a population and compute the same statistic t (say $\overline{x}$) for each one of them. Let $t_1, t_2, \ldots, t_k$ be the sample value of the statistic for these samples. Now, each one of these values can be used to test the null hypothesis Ho. Some values might cause Ho to be rejected, while others might cause Ho to be accepted. This sample statistic $t_1, t_2, \ldots, t_k$ can therefore be separated into two mutually exclusive regions. While one leads to Ho's rejection, the other leads to Ho's acceptance. The region that leads to Ho's rejection is referred to as the 'critical region', while that which leads to Ho's acceptance is referred to as the 'acceptance region'. The test statistic's critical value is the value for which the test simply rejects the null hypothesis at the given significance level.

2.4.5 Confidence Interval

Due to random sampling error, it is impossible to know the exact value of the population parameter using only the information in the sample. However, we can generate a collection of values (or set of values) containing the population parameter with a pre-specified probability using the data from a random sample. Such a pre-specified probability, which says that the population parameter is contained in this set, is termed as the 'confidence level'. Such a set is called a 'confidence set'. The population parameter's confidence set represents all possible values of the population parameter lying between the lower limit and the upper limit. This is called the 'confidence interval'.

2.5 ESTIMATION METHODS

There are three methods of estimating the parameters of a population regression function. These are: (i) method of least squares, (ii) method of maximum likelihood and (iii) method of moments.

2.5.1 Method of Least Squares

Consider a system of m linear equations with n unknowns as follows:

$$a_{11}x_1 + a_{12}x_2 + \quad + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \quad + a_{2n}x_n = b_2$$

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$$a_{m1}x_1 + a_{m2}x_2 + \quad + a_{mn}x_n = b_m \qquad \ldots(2.5)$$

where, a_{ij} (i = 1,2, …., m; j = 1, 2, ….., n) and b_i (i = 1, 2, ….., m) are constants and x_j (j = 1, 2, ….n) are the 'n' unknowns. The system of equations (2.5) will not have a unique solution if the number of equations is more than the number of unknowns ($m > n$). In such a case, we have to find a set of values of x_j (j = 1, 2,… n) which best satisfies the equations in (2.5). This means, such a set of solution should be as close as possible to the actual values of the unknowns. This is achieved by a method known as the 'method of least squares'.

Note that for the equations (2.5) [which are like $y_i = a + bx_i + e_i$], for a set of values x_1, x_2, ……, x_n, $e_i = a_{i1}x_1 + a_{i2}x_2 + \quad + a_{in}x_n - b_i$, i = 1, 2, … m are the residues. Hence, $S = \sum_{i=1}^{m} e_i^2$ is the 'sum of the squares of the residues'. The least squares principle states that the best values of the unknown are those for which the 'sum of the squares of the residues' is the least. Hence, to compute the values of x_j (j = 1, 2, … n) such that S is minimum, we need to have $dS/dx_j = 0$, j = 1, 2, …., n. That is,

$$dS/dx_1 = dS/dx_2 = = dS/dx_n = 0 \qquad \ldots(2.6)$$

The conditions in (2.6) gives n equations involving n unknowns. These equations are called normal equations. Solving these n normal equations with n unknowns we get a unique best solution. Let us take an example to understand this better. Say we want to fit a straight line $y = a + bx$ for a set of a given points (x_i, y_i), i= 1, 2,...,n. For any x_i the estimated value is $\hat{y}_i = a + bx_i$. Now, S, the Sum of the squares of the residuals is:

$S = \sum_{i=1}^{n}(\hat{y_i} - y_i)^2 = \sum_{i=1}^{n}(a + b\, x_i - y_i)^2$

Applying the method of least squares to calculate *a* and *b*, requires us to have:

$$dS/da = 0, dS/db = 0,\ and\ d^2S/da^2 > 0, d^2S/db^2 > 0, \qquad \ldots(2.7)$$

The conditions (2.7) give two normal equations as follows:

$$\sum_{i=1}^{n} y_i = na + b\sum_{i=1}^{n} x_i$$

and

$$\sum_{i=1}^{n} x_i\ y_i = a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} x_i^2 \qquad \ldots(2.8)$$

Solving the two equations in (2.8), we can find the values of *a* and *b*. If $\boldsymbol{a}^*$and $\boldsymbol{b}^*$ are the calculated values of *a* and *b* respectively, then the best fitted line will be

$y = \boldsymbol{a}^* + \boldsymbol{b}^*\ \boldsymbol{x}$

2.5.2 Method of Maximum Likelihood

Let $X_1, X_2, \ldots\ldots X_n$ be *n* random variables from a population with the probability density function (p.d.f.) $f(X, \emptyset)$ where $\emptyset$ is the unknown parameter. The probability is a function of sample values $X_1, X_2 \ldots X_n$ and the parameters $\emptyset_1, \emptyset_2 \ldots \emptyset_k$. The likelihood function is expressed as:

$L(\emptyset) = L(X_1, X_2, \ldots, X_n, \emptyset_1, \emptyset_2 \ldots \emptyset_k)$

The maximum likelihood estimator (MLE) is derived by using the principal of maximum and minimum to the above likelihood function as:

$$\frac{dL}{d\emptyset} = 0 \text{ and } \frac{d^2L}{d\emptyset^2} < 0 \qquad \ldots(2.9)$$

Since $L > 0$, $log\ L(\emptyset)$ attains its maximum for the same value of $\emptyset$ *as* $L(\emptyset)$. Thus, the maximum likelihood estimator of $\emptyset$ is derived by differentiating the $log\ L(\emptyset)$, i.e., the value of $\emptyset$ for which:

$$\frac{dlog(L(\emptyset))}{d\emptyset} = 0 \text{ and } \frac{d^2 log(L(\emptyset))}{d\emptyset^2} < 0$$

First let us consider the case where all x_i's are discrete. This means, P[$\emptyset$, ($X_1 = x_1, X_2 = x_2 \ldots\ldots X_n = x_n$) is a function of $\emptyset$ and $x_1, x_2 \ldots x_n$. For fixed ($x_1, x_2 \ldots x_n$), we therefore have:

$L(\emptyset)$= P[$\emptyset$, $(X_1 = x_1, X_2 = x_2, \ldots\ldots X_n = x_n)$]

=P[$\emptyset, X_1=x_1$].P[$\emptyset, X_2=x_2$]$\ldots\ldots$P[$\emptyset, X_n=x_n$].[since $x_1, \ldots, x_n$ are independent]

$= \prod_{i=1}^{n} P\,\emptyset(Xi = xi) = \prod_{i=1}^{n} f\,\emptyset(xi)$

where *f* is the common 'probability function' (either a mass function for discrete variables or a density function for continuous variables). A value $\breve{\theta}$ of θ is now the maximum likelihood estimator, if $L(\breve{\theta}) \geq L(\theta)$ for all θ. In most cases, we take Log $L(\theta)$, so that further treatments become easier since Log is a one-to-one function. The equation, $\frac{d(\log L(\theta))}{d\,L(\theta)} = 0$, is known as the likelihood equation.

We can similarly consider the case where all x_i's are continuous where $L(\theta)$ is the joint *p.d.f* of ($x_1, x_2 \ldots . x_n$), $dx_1.dx_2 \ldots\ldots dx_n$

$\therefore\ L(\theta) = f(x_1, x_2 \ldots . x_n), dx_1.dx_2 \ldots\ldots dx_n$

$= \prod_{i=1}^{n} f\ \emptyset(xi)$ (since X_i's are independent)

A likelihood function is thus the product of individual probability functions. Let us now consider the MLEs of some standard distributions.

Binomial Distribution: Let X_i be B(m, p) with pmf given by $f_p(x_i) = mCx_i\, p^{x_i}\, (1-p)^{m-x_i}$, i.e., each x_i is an independent binomial variable with (m, p) as its parameters. Therefore, the likelihood function of x_i is given by:

$L(x_i) = \prod_{i=1}^{n} fp\ (x_i)$

$= \prod_{i=1}^{n} mCx_i\, p^{x_i}\, (1-p)^{m-x_i}$

$= (\prod_{i=1}^{n} mCx_i) p^{\Sigma xi} (1-p)^{nm - \Sigma_{i=1}^{n} xi}$

Taking log on both sides, we get:

$$\text{Log } L(x_i) = \text{Log } (\prod_{i=1}^{n} mCx_i) + \Sigma x_i \log p + (nm - \Sigma x_i) \log(1-p) \quad \ldots (2.10)$$

Differentiating equation (2.10) with respect to p we get

$$\frac{d \log L(x_i)}{dp} = \frac{\Sigma x_i}{p} + \frac{nm - \Sigma x_i}{(1-p)}.(-1)$$

$$\frac{L(\acute{x\imath})}{L(xi)} = \frac{\Sigma x_i}{p} - \frac{nm - \Sigma x_i}{(1-p)}$$

$$= \frac{\Sigma xi - p\Sigma x_i - mnp + p\Sigma x_i}{p\ (1-p)}$$

$$= \frac{\Sigma x_i - mnp}{p\ (1-p)}$$

Equating the first order condition to zero we get

$$\frac{\Sigma x_i - mnp}{p\ (1-p)} = 0$$

or $\quad p = \frac{\Sigma x_i}{mn} = \hat{p}$

Therefore, $\hat{p} = \frac{\Sigma x_i}{mn}$ is the maximum likelihood estimator of p.

Poisson Distribution: Now, let $x_1, x_2 \ldots . x_n$ be a sample of n observations drawn from a Poisson distribution having parameter m. The probability mass function (pmf) of x is given by:

$$fm\ (x) = \frac{e^{-m}.m^{x}}{x!} \ (x = 0,1,2, \ldots . \infty)$$

$$\therefore\ L(x) = \prod_{i=1}^{n} fm\ (x_i)$$

$$= \prod_{i=1}^{n} \frac{e^{-m}.m^{x_i}}{x_i!}$$

Taking log on both sides, we get:

$$Log\ L(x) = \log(\prod_{i=1}^{n} e^{-m}) + \sum x_i \log m - \sum_{i=1}^{n} \log x_i!$$

$$= \log e^{-nm} + \sum x_i \log m - \sum_{i=1}^{n} \log x_i! \qquad \ldots (2.11)$$

Differentiating equation (2.11) with respect to *m* we get,

$$\frac{\partial Log\ L(x)}{\partial m} = -n + \frac{\Sigma x_i}{m}$$

Equating the first order condition (FOC) to zero we get:

$$-n + \frac{\Sigma x_i}{m} = 0$$

$$\frac{\Sigma x_i}{m} = n$$

$$\frac{\Sigma x_i}{n} = m$$

$$\bar{x} = m = \widehat{m}$$

$\therefore \widehat{m} = \overline{x}$ is the MLE of the Poisson parameter *m.*

Normal Distribution: Let us now derive the MLE of the parameter σ^2 in a normal distribution with $N(\mu,\sigma^2)$. Let us first assume that the parameter μ is known. Now, the probability density function (*pdf*) of the normal variable *X* is:

$$f(x;\mu,\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\frac{-(x-\mu)2}{2\sigma^2}} - \infty < x < \infty$$

The likelihood function is given by

$$L(x;\mu,\sigma^2) = -n\ \log\sigma - \frac{n}{2}\log(2\pi) - \frac{\Sigma(x_i-\mu)^2}{2\sigma^2}$$

$$= -\frac{n}{2}\log\sigma^2 - \frac{n}{2}\log(2\pi) - \frac{\Sigma(x_i-\mu)^2}{2\sigma^2} \qquad \ldots (2.12)$$

Differentiating equation (2.12) with respect to σ^2we get,

$$\frac{\partial\ Log\ L(x;\mu,\sigma^2)}{\partial\sigma^2} = \frac{-n}{2\sigma^2} + \frac{\Sigma(x_i-\mu)^2}{2\sigma^4}$$

$\therefore$ Applying the FOC $=$ 0 we obtain

$$\frac{-n}{2\sigma^2} + \frac{\Sigma(x_i-\mu)^2}{2\sigma^4} = 0$$

$$or, \sigma^2 = \frac{\Sigma(x_i-\mu)^2}{n} = \hat{\sigma}^2$$

$$\therefore \hat{\sigma}^2 \text{is the M. L. E. of } \sigma^2$$

Note that $\hat{\sigma}^2 is\ an\ unbasied\ estimator\ of\ \sigma^2$ because:

$$E(\hat{\sigma}^2) = \mathrm{E}\left[\frac{\Sigma(x_i-\mu)^2}{n}\right]$$

$$= \frac{1}{n}\mathrm{E}[\Sigma(x_i-\mu)^2]$$

$= \frac{1}{n}.\mathrm{n}\,\sigma^2 = \sigma^2$

Let us now consider the case where both μ & σ^2 are unknown. The likelihood function is given by:

$$Log\,L = -\frac{n}{2}Log\,\sigma^2 \; -\frac{n}{2}Log(2\pi) - \frac{\Sigma(x_i-\mu)^2}{2\sigma^2} \quad \ldots (2.13)$$

Differentiating equation (2.13) with respect to μ we get:

$$\frac{dLogL}{d\mu} = -\frac{2\Sigma(x_i-\mu)}{2\sigma^2}(-1)$$

$$= -\frac{\Sigma(x_i-\mu)}{\sigma^2}$$

Applying FOC = 0, we get: $\frac{\Sigma(x_i-\mu)}{\sigma^2} = 0$ or $\Sigma x_i = n\mu$

$\therefore \frac{\Sigma x_i}{n} = \mu$ or $\overline{x} = \mu = \hat{\mu}$

$\overline{x}$ is the MLE of μ. Likewise:

$\hat{\sigma}^2 = \frac{\Sigma(x_i-\mu)^2}{n} = s^2$ [sample variance] is the MLE of σ^2.

Properties of MLE: Based on the above discussion, we can list the properties of MLE as follows:

i) The MLE is consistent, efficient and sufficient.

ii) The MLE is not necessarily unbiased. But when the MLE is biased, by a slight modification, it can be converted in to an unbiased estimator.

iii) The MLE tends to be distributed normally for large sample.

iv) The MLE is invariant under functional transformation. This means if *T* is an MLE of *ø*, and *g(ø)* is a *function* of *ø*, then *g(T)* is an MLE of *g(ø)*.

2.5.3 Method of Moments

When we fit a particular observed distribution based on a sample of observations, our objective is to see whether the observed distribution is a random sample from its parent distribution. Therefore, before estimating the parameters of a theoretical distribution from the observed data, the first step is the fitting of a distribution. This is particularly the case when the parameter values are assumed or known. In such cases, the method of moments is an alternative method for estimating the parameters of a parent or theoretical distribution. In this method, the first *K* moments of the theoretical distribution expressed in terms of parameters are equated to their corresponding sample moments. In the second step, the *K* equations are solved to obtain the required population estimates.

Let $X_1, X_2 \ldots\ldots X_n$ be a random sample of size *n* drawn from a population. The k^{th} order raw moment is given by: $m_k = \frac{\sum_{i=1}^{k} X_i^k}{n}$. Therefore, E(m_k) = μ_k = 1/*n**E(X_i^k). You should note that μ_k will be a function of the underlying parameters in most cases. We produce K equations by solving which we can get estimates of K unknown parameters. The moments based on powers of X provide a source of parameter information. Let us now see some application of this method of estimation to some standard distributions.

Let $X_1, X_2, \ldots, X_n$ be Bernoulli random variables with parameter p. What is the moment's estimator of p? For Bernouli distribution, the first moment about zero is: $E(X_i) = p$. Hence, we have to derive the method of moments estimator for the only parameter p for which only one equation is required. Hence, equating the first theoretical moment about zero with the corresponding sample moment, we get: $p = \frac{1}{n}\sum_{i=1}^{n} x_i$. Therefore, the moment estimator for p is: $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} x_i$.

Now, consider a sample of drawn from n *i.i.d.* random variables, $X_1, X_2, \ldots, X_n$, with mean μ and variance σ^2. What are the moments estimates of μ and σ^2? We evidently need two moments now. The first moment about zero of a normal distribution is $E(X_i) = \mu$. The second moment about zero of a normal distribution is $E(X_i^2) = \sigma^2 + \mu^2$. We thus have the two equations required to derive the estimates of two parameters. Equating the first theoretical moment about zero with the corresponding sample moment, we get:

$$E(X_i) = \mu = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \ldots (2.14)$$

Equating the second theoretical moment about zero with the corresponding sample moment, we get:

$$E(X_i^2) = \sigma^2 + \mu^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 \quad \ldots (2.15)$$

Equation (2.14) gives the moments estimator of mean μ, the sample mean, as:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$$

Likewise, from equation (2.15), we get:

$$\sigma^2 = \frac{1}{n}\sum X_i^2 - \mu^2$$

$$= \frac{1}{n}\sum X_i^2 - \bar{x}^2$$

$$= \frac{1}{n}\sum (X_i - \bar{x})^2$$

Hence, the method of moments estimator of variance σ^2 is given by

$$\hat{\sigma}^2 = \frac{1}{n}\sum (X_i - \bar{x})^2$$

Check Your Progress 2

1) When is a hypothesis said to be 'complete'? By what alternative name is it called?

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2) Distinguish between a 'one-tailed test' and a 'two-tailed test'.

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3) Distinguish between 'type-I error' and 'type-II error'. How is it related to the 'level of significance'?

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4) State the principle behind the 'method of least squares'. When will it have a 'unique solution'?

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5) State the properties of a maximum likelihood estimator.

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2.6 LET US SUM UP

In this unit we discussed the classical statistical inference theory. Classical statistical inference consists of two parts: estimation and hypothesis testing. Estimation is the process of estimating a population parameter using the data drawn as a sample from the population. Hypothesis testing is the process of evaluating whether to accept or reject a hypothesis or assumption made about the value of a population parameter based on sample data. Under estimation, the unit has explained the concepts of point estimation and interval estimation. Properties of a good estimator are also discussed. Different estimation methods such as Method of Least Squares, Method of Maximum Likelihood and Method of Moments are discussed with illustrations.

2.7 KEY WORDS

Point Estimation : Specific value of a 'statistic' where the term statistic means a function of sample observations (e.g., mean, standard deviation).

Interval Estimation : It refers to the range within which the estimated value of the parameter lies. Thus it refers to a specific interval which depends upon the standard error and the level of significance. If T_1 and T_2 are two functions based on sample observations such that '$P(T_1 \leq \emptyset \leq T_2) = (1 - \alpha)$', (where α is the level of significance), then the interval (T_1, T_2) is termed as an 'interval estimate' or a confidence interval.

Independent and Identically Distributed (i.i.d.) : A collection of random variables is said to be independent and identically distributed if they fulfil two conditions: (i) each random variable has the same distribution (say, for example, normal or Poisson), and (ii) all the random variables are independent (it implies that knowledge of the value of one variable does not provide any clue about the value of the other variables). In symbols, $\boldsymbol{P(AB) = P(A) \times P(B)}$.

Law of Large Numbers (LLN) : Tells us how sample mean can be considered a true representative estimate of population mean with the sample size getting infinitely large. There are two versions of this law, viz., strong LLN and weak LLN. The SLLN says that if A_n is the sample mean and μ is the population mean, then $\lim_{n\to\infty} P(|A_n - \mu| = 0) = 1$]. The WLLN, on the other hand, tells us that for any $\alpha > 0$, $\lim_{n\to\infty} P(|A_n - \mu| < \alpha) = 1$ or $\lim_{n\to\infty} P(|A_n - \mu| > \alpha) = 0$.

Central Limit Theorem : This result says that if X_i, i = 1, 2, ..., n are i.i.d. N(μ, σ^2), then, as $n \longrightarrow \infty$, $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$. Further, the standardised normal variate, $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is N (0, 1).

Method of Least Squares : The least squares principle states that the best values of the unknown are those for which the 'sum of the squares of the residues' is the least.

Likelihood Function : If $f(X, \theta)$ is the probability function, then its likelihood function is defined as: $L(\theta) = \prod_{i=1}^{n} f\,\theta(xi)$.

Maximum Likelihood Estimation (MLE)	:	The MLE method maximises the likelihood function to obtain an estimate of θ as a value $\breve{\theta}$ such that $L(\breve{\theta}) \geq L(\theta)$ for all θ.
Method of Moments	:	The method first equates the moments of the theoretical distribution (expressed in terms of parameters) with their corresponding sample moments. It then proceeds to solve the equations to obtain the estimates of the population parameters.

2.8 SOME USEFUL BOOKS

Bhaumik, Sankar (2017), *Principles of Econometrics: A Modern Approach Using E-views*, Oxford University Press.

Rasch Dieter, Schott Dieter (2018), *Mathematical Statistics*, Wiley.

2.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) A parameter relates to the unknown in the population (e.g. mean, variance). A statistic relates to the estimated mean or variance on the basis of a sample drawn from a population. A statistic is a function of sample observations.

2) A point estimate of a parameter is a specific value of a 'statistic'. Any particular estimate obtained from a sample is a 'point estimate'. If T_1 and T_2 are two estimates such that 'P($T_1 \leq \emptyset \leq T_2$) = 1-$\alpha$' (where α is the level of significance), the interval (T_1, T_2) is termed as an 'interval estimate'. It is also called a 'confidence interval'.

3) While the estimator will be the same for multiple samples, the estimate will vary from sample to sample. The distribution of many such point estimates, obtained from different samples, is called the 'sampling distribution' of the 'statistic'.

4) (i) unbiasedness, ii) consistency, iii) efficiency and iv) sufficiency.

5) (i) $\lim_{n\to\infty} [E(T) - \emptyset] = 0$ and (ii) $\lim_{n\to\infty} Var(T) = 0$.

6) If $\boldsymbol{A_n}$ is the sample mean for any n i.i.d. random variables with mean μ, then for the WLLN, then any positive number $\boldsymbol{\alpha}$, we will have $\lim_{n\to\infty} P(|A_n - \mu| < \alpha) = 1$. For the SLLN, we will have $\lim_{n\to\infty} P(|A_n - \mu| = 0) = 1$.

Check Your Progress 2

1) A hypothesis is said to be complete if all the parameters of the population from which the sample (to test the hypothesis) is drawn are completely specified. Such hypotheses are also called 'simple hypotheses'. This is in

contrast to a 'composite hypothesis', which refers to a hypothesis in which one or more of the parameters are either not exactly specified or not specified at all.

2) If the alternative hypothesis is specified as H_1: $\mu \neq \mu_o$ then the test is called the two-tailed test. If the alternative hypothesis is one-directional like H_1: $\mu < \mu_0$ or $\mu > \mu_0$ then the test is called a one-tailed test. It is either a left tailed test or a right tailed test depending on whether the inequality is < or > in H_1.

3) Rejecting a null hypothesis, when it is actually true, is termed as type-I error. The opposite, i.e., 'accepting an alternative hypothesis when it is actually false' is termed as type-II error. Between the two, the type-I error is considered more grave. Hence, we try to ensure that this is kept as low as possible. This level at which we keep the test is called the 'level of significance', usually denoted by $\propto$, and this is kept at either 1 percent or 5 percent level. This is the maximum permissible level of committing the type-I error.

4) It states that the best values of the unknown are those for which the 'sum of the squares of the residuals', $S = \sum_{i=1}^{m} e^2{}_i$, is the least. The method of least squares gives a unique solution when the number of the normal equations is exactly equal to the number of unknown parameters or the coefficients of the regression model.

5) The properties of a maximum likelihood estimator are as follows: (a) It is consistent, efficient and sufficient, (b) it is not necessarily unbiased but can be made unbiased, (c) it tends to be distributed normally for large samples, and (d) it is invariant to functional transformation.

UNIT 3 REVIEW OF MATRIX ALGEBRA*

Structure

3.0 OBJECTIVES

After going through this unit, you will be able to

- carry out various matrix operations such as multiplication and inverse;
- find out the determinant and rank of a matrix;

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- define eigenvalue and eigenvector and compute them;
- identify some special matrices which are relevant to econometrics;
- compute Kronecker product and vec-operator; and
- calculate Jacobian matrix, Gradient matrix and Hessian matrix.

3.1 INTRODUCTION

Matrix algebra is an integral part of econometrics. It not only helps in presentation of multiple variables in a concise form, but also facilitates various mathematical operations. Further, matrix algebra allows us to generalise results from a small number of variables to larger ones.

You have already studied matrix algebra in the course MEC 203: Quantitative Methods. In the present Unit we provide an overview of certain concepts which are usually applied in econometrics. Our emphasis is on certain special matrices and their properties. Although the coverage of the topics in the present Unit is limited, we have attempted to provide examples of the concepts for easier comprehension.

3.1 BASIC NOTATIONS

Let us begin with the definition of a matrix. As you know, matrix is an array of numbers. A matrix is denoted by an upper case alphabet in boldface (e.g., **A**) and its (i, j)th element (the element at the ith row and jth column) is denoted by corresponding lower case alphabet with subscript i,j (e.g., a_{ij}). The dimension or order of a matrix is given by m×n, when the matrix has m rows and n columns. For example, matrix A can be expressed as

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

If m = n then the matrix is called a 'square matrix'. An n×1 matrix is an n-dimensional column vector. Similarly, an 1×n matrix is a row vector. Every vector will be denoted by a lower-case alphabet in boldface (e.g., **a**) and its element is denoted by corresponding lower case alphabet with subscript i (e.g., a_i). An 1×1 matrix is just a scalar. For a matrix **A** its ith column is denoted by a_i.

Alternatively, $\boldsymbol{A} = [a_1, \quad a_2 \quad \ldots \quad a_n]$

where $\boldsymbol{a_i} = [a_{1i}, \quad a_{2i} \quad \cdots \quad a_{mi}]^T \quad \forall i = 1, \ldots n$

A matrix is said to be diagonal if its off-diagonal elements (i.e., $a_{ij}, i \neq j$) are all zero and at least one of its diagonal elements is non-zero, i.e., $a_{ii} \neq 0$for some i = 1, 2, …, n. A diagonal matrix whose diagonal elements are all one is an identity matrix denoted as I; we also write the n × n identity matrix as I_n.

A matrix **A** is said to be 'lower-triangular' if $a_{ij} = 0$ for $i < j$. Similarly, a matrix is said to be 'upper-triangular' if $a_{ij} = 0$ for $i > j$. We denote **0** (null matrix) as a matrix whose all elements are zero. The transpose of a m × n matrix

$\boldsymbol{A} = (a_{ij})$ is n × m matrix. Thus, $\boldsymbol{A}' = (a_{ji})$. Note that the transpose of a matrix can be denoted by the super-script "T". Thus, for the matrix **A**, the notations $\boldsymbol{A'}$ and $\boldsymbol{A^T}$ are used interchangeably.

3.3 MULTIPLICATION OF MATRICES

If **A** is m × n matrix and **B** is p × q matrix then the product of **A** and **B** is defined as

$\boldsymbol{AB} = \left(C_{ij}\right)_{m\times q}$ where $C_{ij} = \left(\sum_{j=1}^{n} a_{ij} b_{jk}\right)$ provided n = p.

The dimension of **AB** is given by m × q.

When the order of **A** and **B** are such that **AB** is defined, we say that **A** and **B** are conformable. For a real-valued square matrix **A**, we say that **A** is 'orthogonal' if

$$\boldsymbol{A'A} = \boldsymbol{AA'} = \boldsymbol{I} \qquad \dots (3.1)$$

For any two matrices **A** and **B** for which **AB** is defined,

$$(\boldsymbol{AB})^{\boldsymbol{T}} = \boldsymbol{B^T A^T} \qquad \dots (3.2)$$

If we multiply a square matrix **A** by itself, that is **AA**, we obtain $\boldsymbol{A^2}$**.**

Let us consider an example of matrix multiplication.

Example 3.1:

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \; \boldsymbol{B} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

Hence, $\boldsymbol{AB} = \begin{pmatrix} a_{11}.b_{11} + a_{12}.b_{21} & a_{11}.b_{12} + a_{12}.b_{22} \\ a_{21}.b_{11} + a_{22}.b_{21} & a_{21}.b_{12} + a_{22}.b_{22} \end{pmatrix}$

Let us consider one more example.

Example 3.2:

If $= \begin{pmatrix} 1 & 2 & 0 \\ 4 & 5 & 6 \end{pmatrix} \qquad \boldsymbol{B} = \begin{pmatrix} 4 & 0 \\ 5 & 2 \\ 6 & 1 \end{pmatrix}$. Find **AB.**

$$\boldsymbol{AB} = \begin{pmatrix} 1.4 + 2.5 + 0.6 & 1.0 + 2.2 + 0.1 \\ 4.4 + 5.5 + 6.6 & 4.0 + 5.2 + 6.1 \end{pmatrix} = \begin{pmatrix} 14 & 4 \\ 77 & 16 \end{pmatrix}$$

If **AB = BA**, then we say that both matrices commute.

Any matrix **B** satisfying $B^{k-1} \neq 0$ and $B^k = 0$ is said to be 'nilpotent matrix' of index k. The matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is nilpotent of index 2.

The following are a few theorems related to matrix multiplication (we do not provide the proofs here; you can refer to the course MEC 203).

(i) $(\boldsymbol{A} + \boldsymbol{B})^2 = (\boldsymbol{B} + \boldsymbol{A})^2$

(ii) $(\boldsymbol{A} + \boldsymbol{B})^2 = \boldsymbol{A}^2 + \boldsymbol{AB} + \boldsymbol{BA} + \boldsymbol{B}^2$

(because, in general, $\boldsymbol{AB} \neq \boldsymbol{BA}$)

(iii) $(\boldsymbol{A} + \boldsymbol{B})^2 = (\boldsymbol{A} + \boldsymbol{B})(\boldsymbol{A} + \boldsymbol{B})$

(iv) $\boldsymbol{A'A} = \boldsymbol{0}$ iff **A = 0**

(v) **AB = 0** iff $\boldsymbol{A}'\ \boldsymbol{AB} = \boldsymbol{0}$

(vi) **AB = AC** iff $\boldsymbol{A}'\boldsymbol{AB} = \boldsymbol{A}'\boldsymbol{AC}$

(vii) **(AB) C** = **A (BC)** for conformable matrices **A, B, C**

(viii) **A(B + C)** = AB + AC for conformable matrices **A, B, C**

(ix) $(\boldsymbol{AB})' = \boldsymbol{B}'\boldsymbol{A}'$

(x) $(\boldsymbol{ABC})' = \boldsymbol{C}'\boldsymbol{B}'\boldsymbol{A}'$

3.4 DETERMINANT AND TRACE OF A MATRIX

Given a square matrix **A**, let A_{ij} denote the sub-matrix obtained from **A** by deleting its i^{th} row and j^{th} column. The determinant of **A** is denoted by the symbol $\det(\boldsymbol{A})$. In some texts, you can see that determinant is denoted by the symbol Δ or $|\boldsymbol{A}|$.

We can find out the determinant of a matrix **A** by the following formula:

$$det(\boldsymbol{A}) = \sum_{i=1}^{m}(-1)^{i+j}a_{ij}\, det(A_{ij}) \qquad \ldots (3.3)$$

for any j = 1, …n where $(-1)^{i+j}$. Note that $det(A_{ij})$ is called the *cofactor* of a_{ij}. This definition is based on the cofactor expansion along the j^{th} column. Equivalently, we can define the determinant by using the cofactor expansion along the i^{th} row.

$$det(\boldsymbol{A}) = \sum_{j=1}^{n}(-1)^{i+j}a_{ij}\, det(A_{ij}) \text{ for any i = 1…n.} \qquad \ldots (3.4)$$

The determinant of a scalar is the scalar itself. The determinant of a 2 × 2 matrix **A** is given by

$|\boldsymbol{A}| = (a_{11}a_{22} - a_{12}a_{21})$

You can find out the determinant of a higher order matrix by using its *cofactors*.

Example 3.3:

$\boldsymbol{A} = \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$. Find $|\boldsymbol{A}|$.

$|\boldsymbol{A}| = 2 \times 5 - 1 \times 4 = 6.$

A square matrix is singular if its determinant is equal to zero. It is called singular because its inverse cannot be calculated. A square matrix with non-zero determinant is said to be 'non-singular' (implies, the matrix has an 'inverse' or the matrix is 'invertible').

Clearly, $det(\boldsymbol{A}) = det(\boldsymbol{A}')$

For a scalar c and an n × n matrix **A**

$det(c\boldsymbol{A}) = c^n\, det(\boldsymbol{A})$

You should note that the determinant of a square matrix with a column (or row) of zeros, is zero. Also, the determinant of a diagonal or triangular matrix is simply the product of all diagonal elements. It can also be shown that the determinant of the product of two square matrices of the same size is the product of their determinants.

$det(\boldsymbol{AB}) = det(\boldsymbol{A})\, det(\boldsymbol{B}) = det(\boldsymbol{BA})$ … (3.5)

Also, for an m × m matrix **A** and n × n matrix **B**

$det(\boldsymbol{A} \otimes \boldsymbol{B}) = det(\boldsymbol{A})^m\, det(\boldsymbol{B})^n$ … (3.6)

The symbol ⊗ is the Kronecker product, which we will discuss later in Section 3.10.

If **A** is orthogonal matrix, we know that $\boldsymbol{AA'} = \boldsymbol{I}$**.**

Therefore,

$det(\boldsymbol{I}) = det(\boldsymbol{A'A}) = [det(\boldsymbol{A})]^2$ … (3.7)

As the determinant of an identity matrix is one, the determinant of an orthogonal matrix must be either 1 or –1.

The *trace* of a square matrix is the sum of its diagonal elements, i.e.,

trace $(\boldsymbol{A}) = \sum_i a_{ii} = tr(\boldsymbol{A})$ … (3.8)

For example, trace $(\boldsymbol{I_n}) = n$

Let **A** and **B** be square matrices of the same order. Let λ and μ be scalars. Then

(i) $tr(\boldsymbol{A} + \boldsymbol{B}) = tr(\boldsymbol{A}) + tr(\boldsymbol{B})$

(ii) $tr(\lambda\boldsymbol{A} + \mu\boldsymbol{B}) = \lambda tr(\boldsymbol{A}) + \mu tr(\boldsymbol{B})$

(iii) $tr(\boldsymbol{A}) = tr(\boldsymbol{A}')$

(iv) $tr(\boldsymbol{AA}') = tr(\boldsymbol{A}'\boldsymbol{A}) = \sum_{ij} a_{ij}^2$

(v) $tr(aa') = tr(a'a) = \sum_i a_i^2$ for any vector a.

3.5 INVERSE OF A MATRIX

A non-singular matrix **A** possesses a unique inverse $\boldsymbol{A^{-1}}$ in the sense that $\boldsymbol{AA^{-1}} = \boldsymbol{A^{-1}A} = I$. Note that a singular matrix cannot be inverted.

Given an invertible matrix **A**, its inverse can be calculated as $\boldsymbol{A}^{-1} = \frac{1}{det(\boldsymbol{A})}\boldsymbol{F}'$**.** Here, $\boldsymbol{F}$ is the matrix of the cofactors of **A.** Recall that the (i,j)th element of $\boldsymbol{F}$ is the cofactor $(-1)^{i+j}\, det(A_{ij})$. The matrix $\boldsymbol{F}'$ is known as the *adjoint* of **A.**

Example 3.4:

Find the inverse of matrix **A**.

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

First, find the determinant of **A,** which is $(a_{11}a_{22} - a_{12}a_{21})$

Second, find the matrix of cofactors, which is $\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$.

The inverse of matrix **A** is

$$\boldsymbol{A}^{-1} = \frac{1}{a_{11}a_{22}-a_{12}a_{21}}\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

Properties of the Inverse of a Matrix

For any non-singular matrix **A**

(i) $(\lambda\boldsymbol{A})^{-1} = \frac{1}{\lambda}\boldsymbol{A}^{-1}$ (for any scalar quantity $\lambda \neq 0$)

(ii) $(\boldsymbol{A}^{-1})^{-1} = \boldsymbol{A}$

(iii) $(\boldsymbol{A}^{-1})^{T} = (\boldsymbol{A}^{T})^{-1}$

(iv) $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$ if **B** is non-singular and of the same order as **A**

(v) If **A** and **B** commute and **B** is non-singular then **A** and $\boldsymbol{B}^{-1}$commute.

(vi) For a diagonal matrix **A**, note that $\boldsymbol{A}^{-1}$ is also diagonal with the diagonal elements a_{ii}^{-1}.

3.6 RANK OF A MATRIX

Before proceeding to define the rank of a matrix, we need to define 'row-reduced echelon form of a matrix'. A matrix is said to be in *row echelon form*, if it satisfies the following:

(1) all rows having only zeros at bottom of **A**

(2) all the non-zero rows of **A** are, as the row number increases then the number of zeros at the beginning of the rows also increases.

(3) In order to convert a matrix into row echelon form we can perform two operations on the rows, i.e., scalar multiplication and addition/subtraction. However, it is important to mention that we have to perform the operation on all the elements of a row.

Now let us define the rank of a matrix. The rank of matrix **A** is the maximum number of linearly independent rows in **A**. Rank can also be defined as the maximum number of linearly independent columns in a matrix.

Hence, we can say rank of **A** = rank of $(\boldsymbol{A}^{T})$

The rank of **A** is the total number of non-zero rows in the row echelon form of **A**.

Example 3.5:

Find the rank of matrix **A**.

$$\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 1 & -2 \\ 2 & 1 & 3 & -1 \\ 2 & 0 & 1 & 4 \end{pmatrix}$$

Reduce this matrix to row reduce echelon form by replacing R_2 of matrix by (row 2 – 2 times row 1)

$$\begin{pmatrix} 1 & 2 & 1 & -2 \\ 0 & -3 & 1 & 3 \\ 2 & 0 & 1 & 4 \end{pmatrix}$$

By replacing row 3 of above matrix by (row 3 – 2 times row 1)

$$\begin{pmatrix} 1 & 2 & 1 & -2 \\ 0 & -3 & 1 & 3 \\ 0 & -2 & -1 & 8 \end{pmatrix}$$

By replacing row 3 of above matrix by (3 times row 3 – 2 time

s row 2) we get

$$\begin{pmatrix} 1 & 2 & 1 & -2 \\ 0 & -3 & 1 & 3 \\ 0 & 0 & -5 & 18 \end{pmatrix}$$

This matrix is in echelon form and there is no zero-row. The rows of the matrix are linearly independent. Thus, the rank of the matrix is three.

You should note the following two points: (i) If the matrix **A** is of dimension $m \times n$ where $m \neq n$, then rank of the matrix is given by: rank(**A**) $\leq min\{m, n\}$. (ii) For a square matrix to be invertible, it must be of full rank. In other words, for **A** of n × n dimension, $\boldsymbol{A}^{-1}$ exists if rank (**A**) = n.

Certain Properties of the Rank of a Matrix

(i) rank $\boldsymbol{A} = \boldsymbol{0} \Leftrightarrow \boldsymbol{A} = \boldsymbol{0}$

(ii) rank $(\boldsymbol{I}_n) = n$

(iii) rank $(\lambda \boldsymbol{A})$ =rank (**A**) if $\lambda \neq 0$

(iv) rank of a diagonal matrix = number of non-zero diagonal elements in the matrix

(v) rank $(\boldsymbol{A} + \boldsymbol{B}) \leq$ rank (**A**) + rank (**B**)

(vi) rank $(\boldsymbol{A} - \boldsymbol{B}) \geq$ |rank (**A**) – rank (**B**)|

(vii) rank (**AB**) ≤ min (rank **A**, rank **B**)

Check Your Progress 1

1) Find the inverse of matrix $A = \begin{bmatrix} 2 & 4 & 1 \\ 4 & 3 & 7 \\ 2 & 1 & 3 \end{bmatrix}$.

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2) Find rank of matrix $A = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$

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3.7 PARTITIONED MATRICES

A partitioned matrix is a matrix of the form

$\boldsymbol{Y} = \begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{pmatrix}$ where **A, B, C, D** are matrices.

All these matrices need not be square however **A** and **B** must have same number of rows. Similarly, **C** and **D** have the same number of rows. The column number of **A** and **C** and **B** and **D** are equal respectively.

Suppose, **A** is m × p matrix, **B** is m × q matrix, **C** is n × p matrix, and **D** is n × q matrix. The dimension of **Y** is {(m + n) × (p + q)}.

A square matrix is 'block-diagonal' if it takes the form where all the blocks are square, but not necessarily of the same order.

For any tow matrices **A** and **B** the direct sum $\boldsymbol{A} \oplus \boldsymbol{B}$ is defined as

$$A \oplus \boldsymbol{B} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{B} \end{bmatrix} \qquad \ldots (3.9)$$

In other words, the direct sum of two matrices is a block matrix whose principal diagonals are these two matrices and the off-diagonal matrices are null matrices (need not be of the same order). Thus, a block diagonal matrix is the direct sum of its block. If a square matrix is diagonal then it is block-diagonal but not *vice versa*.

We describe some results pertaining to partitioned matrices.

Let $\boldsymbol{Y_1} = \begin{pmatrix} \boldsymbol{A_1} & \boldsymbol{B_1} \\ \boldsymbol{C_1} & \boldsymbol{D_1} \end{pmatrix} \qquad \boldsymbol{Y_2} = \begin{pmatrix} \boldsymbol{A_2} & \boldsymbol{B_2} \\ \boldsymbol{C_2} & \boldsymbol{D_2} \end{pmatrix}$

Then,

$$\boldsymbol{Y} = \boldsymbol{Y_1}\boldsymbol{Y_2} = \begin{pmatrix} \boldsymbol{A_1} + \boldsymbol{A_2} & \boldsymbol{B_1} + \boldsymbol{B_2} \\ \boldsymbol{C_1} + \boldsymbol{C_2} & \boldsymbol{D_1} + \boldsymbol{D_2} \end{pmatrix}$$

If the submatrices are well-defined, i.e., $\boldsymbol{Y_1}$ be an $(m+n) \times (p+q)$ matrix and $\boldsymbol{Y_2}$ be $(p+q) \times (r+s)$ matrix, then

$$\boldsymbol{Y} = \boldsymbol{Y_1}\boldsymbol{Y_2} = \begin{pmatrix} \boldsymbol{A_1}\boldsymbol{A_2} + \boldsymbol{B_1}\boldsymbol{C_2} & \boldsymbol{A_1}\boldsymbol{B_2} + \boldsymbol{B_1}\boldsymbol{D_2} \\ \boldsymbol{C_1}\boldsymbol{A_2} + \boldsymbol{D_1}\boldsymbol{C_2} & \boldsymbol{C_1}\boldsymbol{B_2} + \boldsymbol{D_1}\boldsymbol{D_2} \end{pmatrix}$$

If $\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{pmatrix}$ then $\boldsymbol{Z}' = \begin{pmatrix} \boldsymbol{A}' & \boldsymbol{C}' \\ \boldsymbol{B}' & \boldsymbol{D}' \end{pmatrix}$ with order of the matrices as follows: $\boldsymbol{A}(m \times p), \boldsymbol{B}(m \times q), \boldsymbol{C}(n \times p)$ and $\boldsymbol{D}(n \times q)$.

Suppose, **A** and **D** are square matrices, but not necessarily of the same order. In that case,

$$tr\begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{pmatrix} = tr\boldsymbol{A} + tr\boldsymbol{B} \qquad \ldots (3.10)$$

If **A** and **D** are non-singular matrices, then

$$\begin{pmatrix} \boldsymbol{A} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{D} \end{pmatrix}^{-1} = \begin{pmatrix} \boldsymbol{A}^{-1} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{D}^{-1} \end{pmatrix}$$

If **B** and **C** are non-singular matrices, then

$$\begin{pmatrix} \boldsymbol{O} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{O} \end{pmatrix}^{-1} = \begin{pmatrix} \boldsymbol{O} & \boldsymbol{C}^{-1} \\ \boldsymbol{B}^{-1} & \boldsymbol{O} \end{pmatrix}$$

and

$$\begin{pmatrix} \boldsymbol{O} & \boldsymbol{B} \\ \boldsymbol{B}^{-1} & \boldsymbol{O} \end{pmatrix}^{-1} = \begin{pmatrix} \boldsymbol{O} & \boldsymbol{B} \\ \boldsymbol{B}^{-1} & \boldsymbol{O} \end{pmatrix}$$

The formula for computing the inverse of a partitioned matrix is as follows:

If **A** and **B** are non-singular matrices

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{A}^{-1} + \boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{F}^{-1}\boldsymbol{C}\boldsymbol{A}^{-1} & -\boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{F}^{-1} \\ -\boldsymbol{F}^{-1}\boldsymbol{C}\boldsymbol{A}^{-1} & \boldsymbol{F}^{-1} \end{bmatrix} \qquad \ldots (3.11)$$

where $\boldsymbol{F} = \boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B}$**.**

Equivalently, if **D** and **G** are non-singular,

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{G}^{-1} & -\boldsymbol{G}^{-1}\boldsymbol{B}\boldsymbol{D}^{-1} \\ -\boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{G}^{-1} & \boldsymbol{D}^{-1} + \boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{G}^{-1}\boldsymbol{B}\boldsymbol{D}^{-1} \end{bmatrix}$$

where $\boldsymbol{G} = \boldsymbol{A} - \boldsymbol{B}\boldsymbol{D}^{-1}\boldsymbol{C}$ provided that the matrix inverses in the expression above are well defined.

3.8 EIGENVALUE AND EIGENVECTOR

Given a square matrix **A**, if $\boldsymbol{AC} = \lambda \boldsymbol{C}$ for some scalar λ and non-zero vector **C**, then **C** is an eigenvector of **A** corresponding to the eigenvalue λ ('Eigen' is a German word that means proper or characteristic). The system $(\boldsymbol{A} - \lambda \boldsymbol{I})C = \boldsymbol{0}$ has a non-trivial solution if an only if

$$det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \boldsymbol{0}. \qquad \ldots (3.12)$$

The above equation is called the characteristic equation of **A**. From this characteristic equation we can solve for the eigenvalues of A. Hence, eigenvalues are also referred to as characteristic roots. Similarly, eigenvectors are called characteristic vectors. You should remember that the eigenvalues and eigenvectors of a real-valued matrix need not be real-valued.

Example 3.6:

Find the characteristic equation and characteristic roots of matrix

$\boldsymbol{B} = \begin{pmatrix} 2 & 5 \\ 2 & 3 \end{pmatrix}$.

For characteristic equation of matrix **B** we should have $det(\boldsymbol{B} - \lambda \boldsymbol{I}) = \boldsymbol{0}$.

Since it is a 2×2 matrix, the $\lambda \boldsymbol{I}$ matrix will be $\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$

Therefore, the matrix $(\boldsymbol{B} - \lambda \boldsymbol{I}) = \begin{pmatrix} 2 & 5 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$

On simplification,

$$\det(\boldsymbol{B} - \lambda \boldsymbol{I}) = \begin{vmatrix} 2-\lambda & 5 \\ 2 & 3-\lambda \end{vmatrix} = (2-\lambda)(3-\lambda) - 10$$

$$= 6 - 3\lambda - 2\lambda + \lambda^2 - 10$$

By equating it to zero, we obtain $6 - 5\lambda + \lambda^2 - 10 = 0$

By re-arranging terms in the above equation, we obtain the characteristic equation as

$\lambda^2 - 5\lambda - 4 = 0$

By solving the above characteristic equation, we obtain the characteristic roots of **B** as follows:

$\lambda^2 - 4\lambda - \lambda - 4 = 0$

or, $\lambda(\lambda - 4) - 1\,(\lambda - 4) = 0$

or, $(\lambda - 1)(\lambda - 4) = 0$

Therefore, the characteristic roots are

$\lambda = 1$ or $\lambda = 4$.

When the order of **A** is n×n, the characteristic equation is an nth-order polynomial in λ and has at most n distinct solutions. These solutions (eigenvalues) are usually complex-valued. If some eigenvalues take the same value, there may exist several eigenvectors corresponding to the same eigenvalue. Given an eigenvalue λ, let $C_1, \ldots, C_k$ be associated eigenvectors. Then,

$$\boldsymbol{A}(a_1c_1 + a_2c_2 + \ldots a_kc_k) = \lambda(a_1c_1 + a_2c_2 + \ldots + a_kc_k)$$

So that any linear combination of the eigenvectors is an eigenvector corresponding to λ.

That is, these eigenvectors are closed under scalar multiplication and vector addition and form the eigenspace corresponding to λ. As such, for a common eigenvalue, we are mainly concerned with those eigenvectors that are linearly independent.

If $\boldsymbol{A}$ $(n \times n)$ possesses n distinct eigenvalues, each eigenvalue must correspond to one eigenvector, which is unique up to scalar multiplication. It is therefore typical to normalize eigenvectors such that the Euclidean norm of these vectors is equal to one. It can also be shown that if the eigenvalues of a matrix are all distinct, their associated eigenvectors must be linearly independent. Let **C** denote the matrix of these eigenvectors and **Λ** denote diagonal matrix with diagonal elements being the eigenvalues of **A**. We can write **AC** = **CΛ** as **C** is non-singular. We have

$$\boldsymbol{C^{-1}AC} = \boldsymbol{\Lambda} \text{ or } \boldsymbol{A} = \boldsymbol{C\Lambda C^{-1}}$$

In this case, **A** is said to be similar to $\boldsymbol{\Lambda}$ (this symbol is the upper case 'lambda' in Greek alphabet). When **A** has n distinct eigenvalues $\lambda_1 \ldots \lambda_n$ (this symbol is the lower case 'lambda') we find that

$$trace(\boldsymbol{A}) = trace(\boldsymbol{C\Lambda C^{-1}}) = trace(\boldsymbol{C^{-1}C\Lambda}) = trace(\boldsymbol{\Lambda})$$

$$det(\boldsymbol{A}) = det(\boldsymbol{C\Lambda C^{-1}}) = det(\boldsymbol{\Lambda})\, det(\boldsymbol{C}).\, det(\boldsymbol{C^{-1}}) = det(\boldsymbol{\Lambda})$$

Let **A** be a $n \times n$ matrix with distinct eigenvalues $\lambda_1 \ldots \lambda_n$. Then,

$det(\boldsymbol{A}) = det(\boldsymbol{\Lambda}) = \prod_{i=1}^{n} \lambda_i$

trace (**A**) = trace $(\boldsymbol{\Lambda}) = \sum_{i=1}^{n} \lambda_i$

when $\boldsymbol{A} = \boldsymbol{C\Lambda C^{-1}}$ we have $\boldsymbol{A^{-1}} = \boldsymbol{C\Lambda^{-1}C^{-1}}$. This shows that the eigenvectors of $\boldsymbol{A^{-1}}$are same as those of **A**, and the corresponding eigenvalues are simply the reciprocals of the eigenvalues of A. similar,

$\boldsymbol{A^2} = (\boldsymbol{C\Lambda C^{-1}})(\boldsymbol{C\Lambda C^{-1}}) = \boldsymbol{C\Lambda^2 C^{-1}}$

The eigenvectors of $\boldsymbol{A^2}$ are the same as those of **A**, and the corresponding eigenvalues are the squares of the eigenvalues of **A**. This result generalizes immediately to $\boldsymbol{A^k}$.

3.9 CERTAIN SPECIAL MATRICES

Let us discuss about symmetric matrices. Let $\boldsymbol{C_1}$ and $\boldsymbol{C_2}$ be two eigenvectors of **A** corresponding to the distinct $\lambda_1$and λ_2 respectively. If **A** is symmetric, then

$\boldsymbol{C_2' A C_1 = \lambda_1 C_2' C_1 = \lambda_2 C_2' C_1}$

As $\lambda_1 \neq \lambda_2$ it must be true that $\boldsymbol{C_2' C_1 = 0}$ so that they are orthogonal. Given linearly independent eigenvectors that correspond to a common eigenvalue, they can also be orthogonalized. Thus, a symmetric matrix is orthogonally diagonalizable, in the sense that

$\boldsymbol{C'AC = \Lambda}$ or $\boldsymbol{A = C\Lambda C'}$

where $\boldsymbol{\Lambda}$ is again the diagonal matrix of the eigenvalues of **A**, and **C** is the orthogonal matrix of associated eigenvectors.

Let **A** (n × n) and **C** (k × k) be non-singular matrices. Then for any n × k matrix **B**,

rank (**B**) = rank (**AB**) = rank (**BC**)

Thus, the rank of a matrix is preserved under non-singular transformations. Similarly, orthogonal transformation does not change the rank of a matrix. For a symmetric matrix **A**, rank (**A**) = rank $(\boldsymbol{\Lambda})$, the number of non-zero eigenvalues of **A**. Moreover, if we can convert **A** to a diagonal matrix, then $det(\boldsymbol{A}) = \prod_{i=1}^{n} \lambda_i$ and trace$(\boldsymbol{A}) = \sum_{i=1}^{n} \lambda_i$ irrespective of the differences in the eigenvalues of **A**.

Note that the above equality of determinant and trace is valid for a non-singular symmetric matrix, whose eigenvalues are non-zero.

A symmetric matrix **A** is said to be positive definite if $\boldsymbol{b'Ab} > 0$for all vectors $\boldsymbol{b} \neq 0$; A is said to be positive semi definite if $\boldsymbol{b'Ab \geq 0}$ for all $b \neq 0$. A positive definite matrix thus must be non-singular, but a positive semi-definite matrix may be singular. Suppose **A** is a symmetric matrix orthogonally diagonalized as $\boldsymbol{C'AC = \Lambda}$. If **A** is also positive semi-definite, then for any $\boldsymbol{b} \neq 0$,

$\boldsymbol{b'\Lambda b = b'(C'AC)b = \tilde{b}'A\tilde{b} \geq 0}$

where $\widetilde{\boldsymbol{b}} = \boldsymbol{cb}$. This shows that $\boldsymbol{\Lambda}$ is also positive simi-definite and all the diagonal elements of $\boldsymbol{\Lambda}$ must be non-negative. You can find out that the converse also holds.

A symmetric matrix is positive definite (positive semi-definite), if and only if its eigenvalues are all positive (non-negative). For a symmetric and positive definite **A**, $\boldsymbol{A^{-1/2}}$is such that $\left(\boldsymbol{A^{-1/2}}\right)\boldsymbol{A^{-1/2}} = \boldsymbol{A^{-1}}$ (Note that $\boldsymbol{A^{-1/2}}$ is the square root of matrix $\boldsymbol{A}$). In particular, by orthogonal diagonalization, we obtain

$$\boldsymbol{A^{-1}} = \boldsymbol{C\Lambda^{-1}C^{-1}} = \left(\boldsymbol{C\Lambda^{-1/2}C'}\right)\left(\boldsymbol{C\Lambda^{-1/2}C'}\right) \qquad \ldots (3.13)$$

So that we may choose $\boldsymbol{A^{-1/2}} = \boldsymbol{C\Lambda^{-1/2}C}$

The inverse of $\boldsymbol{A^{-1/2}}$ is $\boldsymbol{A^{1/2}}$ which is equal to $\boldsymbol{C\Lambda^{1/2}C'}$

It follows that $\boldsymbol{A^{1/2}}(\boldsymbol{A^{1/2}}) = \boldsymbol{A}$ and $\boldsymbol{A^{-1/2}A}\left(\boldsymbol{A^{-\frac{1}{2}}}\right)' = \boldsymbol{I}$

The square root of a positive semi-definite matrix can be found out by the method of 'Cholesky Decomposition'. Given two symmetric and positive definite matrices **A** and **B** if **A – B** is positive semi-definite, then so $\boldsymbol{B^{-1}} - \boldsymbol{A^{-1}}$.

3.9.1 QR Factorization Matrices

If **A** is a real m×n matrix of rank n, then there is exists an m×n semi-orthogonal matrix **Q** (that is, a real matrix satisfying $\boldsymbol{QQ} = \boldsymbol{I_n}$) and a n×n real upper triangular matrix **R** with positive diagonal elements. Such that **A** = **QR**. The result $q_1, q_2, \ldots, q_n$which are columns of **Q** can be obtained by Gram-Schmidt process.

3.9.2 Singular Value Decomposition

Let **A** be real m × n matrix with rank (**A**) = r > 0. Then there exist a m × r matrix **S** such that $\boldsymbol{S'S} = \boldsymbol{I_r}$ and $n \times r$ matrix **T** such that $\boldsymbol{T'T} = \boldsymbol{I_r}$ and a r × r diagonal matrix $\boldsymbol{\Lambda}$ with positive diagonal elements such that $\boldsymbol{A} = \boldsymbol{S\Lambda^{\frac{1}{2}}T'}$since $AA'$is an m × m symmetric (in fact positive semidefinite) matrix of rank r. The **S** and **T** can be found by formula $\boldsymbol{AA'S} = \boldsymbol{S\Lambda}$ and $\boldsymbol{T} = \boldsymbol{A'S\Lambda^{-1/2}}$ respectively.

3.9.3 Idempotent Matrix

The matrix **A** is said to be idempotent if **AA = A.**

Properties of an idempotent matrix are as follows:

(i) If **A** is idempotent and diagonal, matrix then the diagonal elements 0 and 1 only.

(ii) If **A** is idempotent, $\boldsymbol{A'}, \boldsymbol{A^k} (k = 1,2,\ldots)$ and **(I–A)** are also idempotent.

(iii) **A** and **B** are idempotent if and only if $\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{A} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{B} \end{pmatrix}$ is idempotent.

(iv) Eigenvalues of an idempotent matrix are **0** and **1**. But the converse is not true.

(v) For any idempotent matrix **A**, note that rank(**A**) = trace(**A**)

Check Your Progress 2

1) Show that a non-singular matrix **T** exists such that $\boldsymbol{T^{-1}AT = \Lambda}$ (diagonal) if and only if there is a set of n linearly independent vectors, each of which is an eigenvector of **A**.

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2) Let **X** be a matrix of order n × k and rank k.

Consider the matrix $\boldsymbol{M = I_n - X(X'X)^{-1}X'}$.

(a) Show that **M** is symmetric idempotent.

(b) Show that **MX** = 0

(c) Show that rank (**M**) = (n – k)

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3) Let **X** be a matrix of order n × k and rank k, let $\boldsymbol{M = I_n - X(X'X)^{-1}X'}$ and let **V** be positive definite of order n such that $\boldsymbol{X'V^{-1}M = 0}$. Show that

(a) $\boldsymbol{(X'V^{-1}X)^{-1}X'V^{-1} = (X'X)^{-1}X'}$

(b) $\boldsymbol{(X'V^{-1}X)^{-1} = (X'X)^{-1}X'VX(X'X)^{-1}}$

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3.10 KRONECKER PRODUCT AND VEC-OPERATOR

We usually use the Kronecker product in multiple regression models to write the normal equations in a compact form. The symbol ⊗ is used to denote the Kronecker product. Kronecker product is an operation on two matrices that gives us a block matrix.

Suppose we have a $m \times n$ matrix $\boldsymbol{A} = [\boldsymbol{a_1}, \ldots . \boldsymbol{a_n}]$ and a p×q matrix **B**. The Kronecker product $\boldsymbol{A \otimes B}$ is a mp × nq matrix given by

$$\boldsymbol{A} \otimes \boldsymbol{B} = \left[a_{ij}\boldsymbol{B}\right]_{i=1,j=1}^{m,n} = \begin{bmatrix} a_{11}\boldsymbol{B} & a_{12}\boldsymbol{B} & \dots & a_{1n}\boldsymbol{B} \\ a_{21}\boldsymbol{B} & a_{22}\boldsymbol{B} & \dots & a_{2n}\boldsymbol{B} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1}\boldsymbol{B} & a_{m2}\boldsymbol{B} & \dots & a_{mn}\boldsymbol{B} \end{bmatrix} \quad \dots (3.14)$$

Example 3.7:

Find Kronecker product $\boldsymbol{A} \otimes \boldsymbol{B}$ for matrices **A** and **B** given below.

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad \boldsymbol{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

For the above matrices, the Kronecker product is given by

$$\boldsymbol{A} \otimes \boldsymbol{B} = \begin{pmatrix} 1\boldsymbol{B} & 2\boldsymbol{B} \\ 3\boldsymbol{B} & 4\boldsymbol{B} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 3 & 0 & 4 \\ 3 & 0 & 4 & 0 \end{pmatrix}$$

Note that the Kronecker product is a mapping such that

$$\mathbb{R}^{m\times n} \times \mathbb{R}^{p\times q} \to \mathbb{R}^{mp\times nq} \quad \dots (3.15)$$

Further, the Kronecker product of **A** and **B**, i.e., $\boldsymbol{A} \otimes \boldsymbol{B}$ is defined for any pair of matrices **A** and **B** irrespective of their orders. When n = 1 and q = 1, the Kronecker product of two matrices reduces to the Kronecker product of two column vectors $\boldsymbol{a} \in \mathbb{R}^{\boldsymbol{m}}$ and $\boldsymbol{b} \in \mathbb{R}^{\boldsymbol{p}}$

$$\boldsymbol{a} \otimes \boldsymbol{b} = [a_i\boldsymbol{b}]_{i=1}^{m} = \begin{bmatrix} a_1\boldsymbol{b} \\ a_m\boldsymbol{b} \end{bmatrix}$$

The result is a mp ×1 vector. Evidently, the *outer product* of two vectors $\boldsymbol{xy} = \boldsymbol{xyT}$ can be also represented by using the Kronecker product as $\boldsymbol{xy} = \boldsymbol{x} \otimes \boldsymbol{yT}$.

Example 3.8:

Consider the following vectors and find their Kronecker product.

$$\boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}' \qquad \boldsymbol{y} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}'$$

Hence, $\boldsymbol{x} \otimes \boldsymbol{y} = \begin{bmatrix} 1\boldsymbol{y} \\ 2\boldsymbol{y} \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$

The Kronecker product is also known as the *direct product* or *tensor product*.

Kronecker product has the following properties:

(i) The Kronecker product of any matrix and a zero matrix is equal to the zero matrix, i.e., $\boldsymbol{A} \otimes \boldsymbol{O} = \boldsymbol{O} \otimes \boldsymbol{A} = \boldsymbol{0}$.

(ii) $\boldsymbol{\alpha A} \otimes \boldsymbol{\beta B} = \boldsymbol{\alpha\beta}(\boldsymbol{A} \otimes \boldsymbol{B})$, for scalar quantities α and β.

(iii) $\boldsymbol{I_m} \otimes \boldsymbol{I_n} = \boldsymbol{I_{mn}}$ for $\boldsymbol{I_{j\times j}}$ is identity matrix of j × j dimension

(iv) For matrices $\boldsymbol{A_{m\times n}}, \boldsymbol{B_{n\times k}}, \boldsymbol{C_{l\times p}}, \boldsymbol{D_{p\div q}}$, we have $(\boldsymbol{AB}) \otimes (\boldsymbol{CD}) = (\boldsymbol{A} \otimes \boldsymbol{C})(\boldsymbol{B} \otimes \boldsymbol{D})$

(v) For matrices $\boldsymbol{A}_{\boldsymbol{m}\times\boldsymbol{n}}, \boldsymbol{B}_{\boldsymbol{p}\times\boldsymbol{q}}, \boldsymbol{C}_{\boldsymbol{p}\times\boldsymbol{q}}$, we have $\boldsymbol{A} \otimes (\boldsymbol{B} \pm \boldsymbol{C}) = \boldsymbol{A} \otimes \boldsymbol{B} \pm \boldsymbol{A} \otimes \boldsymbol{C}, (\boldsymbol{B} \pm \boldsymbol{C}) \otimes \boldsymbol{A} = \boldsymbol{B} \otimes \boldsymbol{A} \pm \boldsymbol{C} \otimes \boldsymbol{A}$

(vi) $(\boldsymbol{A} \otimes \boldsymbol{B})^{-\boldsymbol{1}} = \boldsymbol{A}^{-\boldsymbol{1}} \otimes \boldsymbol{B}^{-\boldsymbol{1}}, (\boldsymbol{A} \otimes \boldsymbol{B})^{\boldsymbol{T}} = \boldsymbol{A}^{+} \otimes \boldsymbol{B}^{+}$

(vii) $(\boldsymbol{A} \otimes \boldsymbol{B})' = \boldsymbol{A}' \otimes \boldsymbol{B}'$

(vii) rank $(\boldsymbol{A} \otimes \boldsymbol{B})$ = rank (**A**) rank (**B**)

(viii) $det(\boldsymbol{A}_{\boldsymbol{n}\times\boldsymbol{n}} \otimes \boldsymbol{B}_{\boldsymbol{m}\times\boldsymbol{m}}) = (det\,\boldsymbol{A})^m (det\,\boldsymbol{B})^n$

(ix) $tr(\boldsymbol{A} \otimes \boldsymbol{B}) = tr(\boldsymbol{A}) + tr(\boldsymbol{B})$

(x) For matrices $\boldsymbol{A}_{\boldsymbol{m}\times\boldsymbol{n}}, \boldsymbol{B}_{\boldsymbol{m}\times\boldsymbol{n}}, \boldsymbol{C}_{\boldsymbol{p}\times\boldsymbol{q}}, \boldsymbol{D}_{\boldsymbol{p}\times\boldsymbol{q}}$, we have

$$(\boldsymbol{A} + \boldsymbol{B}) \otimes (\boldsymbol{C} + \boldsymbol{D}) = \boldsymbol{A} \otimes \boldsymbol{C} + \boldsymbol{A} \otimes \boldsymbol{D} + \boldsymbol{B} \otimes \boldsymbol{C} + \boldsymbol{B} \otimes \boldsymbol{D}$$

(xi) For matrices $\boldsymbol{A}_{\boldsymbol{m}\times\boldsymbol{n}}, \boldsymbol{B}_{\boldsymbol{p}\times\boldsymbol{q}}, \boldsymbol{C}_{\boldsymbol{k}\times\boldsymbol{l}}$ we have

$$(\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} = \boldsymbol{A} \otimes (\boldsymbol{B} \otimes \boldsymbol{C})$$

3.10.1 Vec-operator (or Vectorization of Matrix)

The vectorization of a matrix $\boldsymbol{A} \in \mathbb{R}^{\boldsymbol{m}\times\boldsymbol{n}}$ denoted as $vec\,(A)$ is a linear transformation which arranges the entries of $\boldsymbol{A} = [a_{ij}]$ as a mn×1 vector by a column.

$vec\,(A) = [a_{11} \ldots a_{m1} \ldots a_{1n} \ldots a_{mn}]^T$

A matrix can be arranged as a row vector by stacking the rows. This is known as row-vectorization of a matrix. It is denoted as $rvec\,(\boldsymbol{A})$. It is defined as

$rvec(A) = [a_{11} \ldots a_{1n} \ldots a_{m1} \ldots a_{mn}]$

Example 3.9:

For the matrix $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ find the vec-operator.

$vec(\boldsymbol{A}) = [a_{11}, a_{21}, a_{12}, a_{22}]^T$ and $rvec(A) = [a_{11}, a_{12}, a_{21}, a_{22}]$

The relationship between column vectorization and row vectorization of a matrix can be expressed as follows:

$rvec(\boldsymbol{A}) = (vec(\boldsymbol{A}^T)^T$

Similarly,

$vec(\boldsymbol{A}^T) = rvec(\boldsymbol{A})^T$

For a given m × n matrix, the two vectors $vec(\boldsymbol{A})$ and $vec(\boldsymbol{A}^T)$ contain the same entries but the order of their entries are different. There is a unique mn×mn permutation matrix that can transform $vec(\boldsymbol{A})$ into $vec(\boldsymbol{A}^T)$. This permutation matrix is known as the *commutation matrix*. It is denoted by K_{mn} and is defined as $K_{mn}vec(\boldsymbol{A}) = vec(\boldsymbol{A}^T)$

Similarly, there is an nm × nm permutation matrix transforming $vec(A^T)$ into $vec(\boldsymbol{A})$. Such a commutation matrix denoted K_{nm}, is defined by

$$K_{nm}vec(A^T) = vec(A)$$

Since this formula holds for any m×n matrix **A,** we have

$$K_{nm}Kk_{mn} = I_{mn} \qquad \text{or, } K_{mn}^{-1} = K_{nm}$$

3.10.2 Properties of Vec-opeartor

(i) If **A** and **B** are matrices with the same order, then

$$vec(\boldsymbol{A} + \boldsymbol{B}) = vec(\boldsymbol{A}) + vec(\boldsymbol{B})$$

(ii) $vec(\alpha\boldsymbol{A}) = \alpha\ vec(\boldsymbol{A})$, for any scalar quantity α

(iii) $vec(\boldsymbol{A}) = vec(\boldsymbol{B})$ implies that **A = B** iff **A** and **B** have the same order.

(iv) For any two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$, $vec\ (\boldsymbol{ab})' = \boldsymbol{b} \otimes \boldsymbol{a}$

3.11 MATRIX DIFFERENTIAL

In Unit 7 of MEC 203: Quantitative Methods you were introduced to the concept of vector differentiation. Let us consider two vectors:

$$\boldsymbol{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \text{ and } \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Let us find out the derivative of $\boldsymbol{y} = \boldsymbol{a}'\boldsymbol{x}$ with respect to $\boldsymbol{x}$. We find that

$$\boldsymbol{a}'\boldsymbol{x} = a_1x_1 + a_2x_2 + a_3x_3 \qquad \dots (3.16)$$

When we take the derivative $\frac{\partial y}{\partial x}$, we actually consider

$$\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \\ \frac{\partial y}{\partial x_3} \end{bmatrix}.$$

On taking partial derivatives of equation (3.16), we obtain

$$\frac{\partial(\boldsymbol{a}'\boldsymbol{x})}{\partial \boldsymbol{x}} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \boldsymbol{a} \qquad \dots (3.17)$$

Note that we started with $\boldsymbol{a}'\boldsymbol{x}$**,** but after differentiation, we obtained $\boldsymbol{a}$**.**

Matrix differential is a generalization of a multivariate function differential. The matrix differential (including matrix partial derivative and gradient) has wide applications in optimization theory and econometrics.

Before proceeding further, let us consider the following notations:

$\boldsymbol{x} = [x_1 \ldots\ldots x_m]^T \in \mathbb{R}R^m$ denotes a real vector variable.

$\boldsymbol{X} = [X_1 \ldots X_n] \in \mathbb{R}^{m\times n}$ denotes a real matrix variable.

The table below shows classification of real functions.

Function Type	Variable $\boldsymbol{x} \in \mathbb{R}^m$	Variable $\boldsymbol{X} \in \mathbb{R}^{m\times n}$
Scalar Function $f \in \mathbb{R}$	$f(\boldsymbol{x}): \mathbb{R}^m \to \mathbb{R}$	$f(\boldsymbol{X}): \mathbb{R}^{m\times n} \to R$
Vector Function $f \in \mathbb{R}^p$	$f(\boldsymbol{x}): \mathbb{R}^m \to \mathbb{R}^p$	$f(\boldsymbol{X}): \mathbb{R}^{m\times n} \to \mathbb{R}^p$
Matrix Function $F \in \mathbb{R}^{p\times q}$	$F(\boldsymbol{x}): \mathbb{R}^m \to \mathbb{R}^{p\times q}$	$F(\boldsymbol{X}): \mathbb{R}^{m\times n} \to \mathbb{R}^{p\times q}$

Let us consider the first case. Here, $f(\boldsymbol{x}): \mathbb{R}^m \to \mathbb{R}$ is a real scalar function with a m×1 real vector $\boldsymbol{x}$ as variable. The function's output is a scalar: for example, $f(\boldsymbol{x}) = trace(\boldsymbol{x}\boldsymbol{x}^{\boldsymbol{T}})$ where $\boldsymbol{x}$ is m ×1 vector.

3.11.1 Jacobian Matrix

The Jacobian of a matrix consists of all first order partial derivates of multivariate function. The row partial derivatives with respect to a m×1 vector $\boldsymbol{x}$ are given by

$$D_X = \frac{\partial}{\partial x^T} = \left[\frac{\partial}{\partial x_1} \cdots\cdots \frac{\partial}{\partial x_m}\right]$$

Similarly, the row partial derivatives with respect to a m×n matrix **X** are given by

$$D_{vecX} = \frac{\partial}{\partial (vecX)^T} = \left[\frac{\partial}{\partial x_{11}} \cdots \frac{\partial}{\partial x_{m1}} \cdots \frac{\partial}{\partial x_{1n}} \cdots \frac{\partial}{\partial x_{mn}}\right]$$

When we collect the partial derivatives for all the rows, we obtain the Jacobian matrix, which is given by

$$D_X = \frac{\partial}{\partial X^T} = \begin{bmatrix} \frac{\partial}{\partial x_{11}} & \cdots & \frac{\partial}{\partial x_{m1}} \\ \vdots & \vdots & \vdots \\ \frac{\partial}{\partial x_{1n}} & \cdots & \frac{\partial}{\partial x_{mn}} \end{bmatrix} \qquad \ldots (3.18)$$

The partial derivative vector resulting from the operation of D_X on a real scalar function $f(\boldsymbol{X})$ with m×1 vector variable $\boldsymbol{X}$ is an 1×m row vector and is given by

$$D_X f(X) = \frac{\partial f(X)}{\partial X^T} = \begin{bmatrix} \frac{\partial f(X)}{\partial x_1} & \cdots & \frac{\partial f(x)}{\partial x_m} \end{bmatrix}$$

The partial derivative matrix

$$D_X f(X) = \frac{\partial f(X)}{\partial X^T} = \begin{bmatrix} \frac{\partial f(X)}{\partial x_{11}} & \cdots & \frac{\partial f(X)}{\partial x_{m1}} \\ \vdots & \vdots & \vdots \\ \frac{\partial f(X)}{\partial x_{1n}} & \cdots & \frac{\partial f}{\partial x_{mn}} \end{bmatrix} \in \mathbb{R}^{n\times m}$$

is known as the Jacobian matrix of the real scalar function $f(X)$, and the partial derivative vector

$$D_{vecX}f(X) = \frac{\partial f(X)}{\partial (vec(X))^T} = \left[\frac{\partial f(X)}{\partial x_{11}} \cdots \frac{\partial f(X)}{\partial x_{m1}} \cdots \frac{\partial f(X)}{\partial x_{1n}} \cdots \frac{\partial f(X)}{\partial x_{mn}}\right]$$

is row partial derivative vector of the real scalar function $f(X)$ with matrix variable *X*.

Example 3.10:

Determine the Jacobian matrix of the function $f: \mathbb{R}^4 \to \mathbb{R}$ defined for all $X = (x_1, x_2, x_3, x_4)$ by $f(X) = \|X\|$

$$f(X) = f(x_1, x_2, x_3, x_4) = \|x\|$$

$$= \|(x_1, x_2, x_3, x_{)4}\| = \sqrt{x_1^2 + x_2^2 + x_3^2 + x_4^2}$$

Hence $D_X f(X) = \left[\frac{\partial\|x\|}{\partial x_1}, \frac{\partial\|X\|}{\partial x_2}, \frac{\partial\|X\|}{\partial x_3}, \frac{\partial\|X\|}{\partial x_4}\right]$

$$= \frac{1}{\sqrt{x_1^2+x_2^2+x_3^2+x_4^2}}[x_1, x_2, x_3, x_4]$$

$$= \frac{1}{\|X\|}(X)$$

3.11.2 Gradient Matrix

Gradient is a term used to define the derivative of a multivariate function. In the case of a single variable, derivative gives us the rate of change in the variable. In the case of multiple variables, there are changes in all the variables. These multiple changes are represented by vectors. As you know, vector a has direction. Thus, the gradient matrix will give us the rate of change of the variables.

The gradient vector operator is denoted by the symbol $\boldsymbol{\nabla}$ (read as 'del' or 'nabla'; it is the inverted upper case letter delta). With respect to a m×1 vector ***X*** it is defined as

$$\nabla_X = \left[\frac{\partial}{\partial x_1} \quad \cdots \quad \frac{\partial}{\partial x_m}\right]^T$$

For a a m×n matrix *X* the gradient operator is defined as

$$\nabla_{vecX} = \frac{\partial}{\partial vecX} = \left[\frac{\partial}{\partial x_{11}} \quad \cdots \quad \frac{\partial}{\partial x_{1n}} \quad \cdots \quad \frac{\partial}{\partial x_{m1}} \quad \cdots \quad \frac{\partial}{\partial x_{mn}}\right]^T$$

The gradient matrix operator with respect to an m×n matrix X is defined as

$$\nabla_X = \frac{\partial}{\partial X} = \begin{bmatrix} \frac{\partial}{\partial x_{11}} & \cdots & \frac{\partial}{\partial x_{1n}} \\ \vdots & \vdots & \vdots \\ \frac{\partial}{\partial x_{m1}} & \cdots & \frac{\partial}{\partial x_{mn}} \end{bmatrix} \qquad \ldots (3.20)$$

The gradient vectors of functions $f(x)$ and $f(X)$ are defined respectively as

$$\nabla_X f(X) = \left[\frac{\partial f(X)}{\partial x_1} \quad \cdots \quad \frac{\partial f(x)}{\partial x_m}\right]^T = \frac{\partial f(X)}{\partial X}$$

$$\nabla_{vec(X)} f(X) = \left[\frac{\partial f(X)}{\partial x_{11}} \quad \cdots \quad \frac{\partial f(X)}{\partial x_{m1}} \quad \cdots \quad \frac{\partial f}{\partial x_{1n}} \quad \cdots \quad \frac{\partial f(X)}{\partial x_{mn}}\right]^T$$

The Gradient matrix of the function f(x) is defined as

$$\nabla_X f(X) = \begin{bmatrix} \frac{\partial f(X)}{\partial x_{11}} & \cdots & \frac{\partial f(X)}{\partial x_{1n}} \\ \vdots & \vdots & \vdots \\ \frac{\partial f(X)}{\partial x_{m1}} & \cdots & \frac{\partial f(x)}{\partial x_{mn}} \end{bmatrix} = \frac{\partial f(X)}{\partial X} \qquad \ldots (3.21)$$

Clearly, there is a relation between gradient matrix and gradient vector of a function $f(X)$.

$$vec\nabla_X f(X) = \nabla_{vecX} f(X) \qquad \ldots (3.22)$$

Further, we can write

$$\nabla_X f(X) = D_X^T f(X)$$

that is, the gradient matrix of a real scalar function $f(X)$ is equal to the transpose of its Jacobian matrix.

Example 3.11:

Let $f(X): R^3 \to R$ be defined as $f(X) = x_1^2 + 2x_1x_2 + 3x_2^2 + 4x_1x_2x_3 + x_3^3$ for

$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Calculate $\nabla_X f(X)$.

The solution is

$$\nabla_X f(X) = \begin{bmatrix} \frac{\partial f(X)}{\partial x} \\ \frac{\partial f(X)}{\partial x_2} \\ \frac{\partial f(X)}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2x_1 + 2x_2 + 4x_2x_3 \\ 2x_1 + 6x_2 + 4x_1x_3 \\ 4x_1x_2 + 3x_3^2 \end{bmatrix}$$

3.11.3 Real Hessian Matrix

Let us Consider a real vector $X \in \mathbb{R}^{m\times 1}$, the second-order derivative of a real function $f(X)$ is known as the Hessian matrix, denoted by H$[f(X)]$ and is defined as

$$H[f(X)] = \frac{\partial^2 f(X)}{\partial X \partial X^T} = \frac{\partial}{\partial X}\left(\frac{\partial f(X)}{\partial X^T}\right) \in \mathbb{R}^{m\times m}$$

which is simply represented as

$$H[f(X)] = \nabla_X^2 f(X) = \nabla_X (D_X f(X)) \qquad \ldots (3.23)$$

In words, Hessian matrix is the gradient matrix of the Jacobian matrix of the function $f(X)$ with respect to *X*.

Hence the (i, j)th entry of the Hessian matrix is defined as

$$[Hf(X)]_{ij} = \left[\frac{\partial^2 f(X)}{\partial X \partial X^T}\right]_{ij} = \frac{\partial}{\partial x_i}\left(\frac{\partial f(X)}{\partial x_j}\right)$$

Therefore, the Hessian matrix is

$$H[f(X)] = \frac{\partial^2 f(X)}{\partial X \partial X^T} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 x_m} \\ \vdots & \vdots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_m \partial x_n} \end{bmatrix} \in \mathbb{R}^{m\times m} \qquad \ldots (3.24)$$

Equation (3.24) shows that the Hessian matrix of a real-valued scalar function $f(X)$ is an m×m matrix that consists of m^2 second-order partial derivatives of $f(X)$ with respect to the entries x_i of the vector variable *X*.

By definition, we can show that the Hessian matrix of a real scalar function $f(X)$ is a real symmetric matrix, i.e.,

$$(H[f(x)])^T = H[f(X)]$$

Because $f(X)$is a second order differentiable continuous function, its second-order derivative is independent of the order of differentiation. It means,

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i} \qquad \ldots (3.25)$$

Example 3.12:

Find Hessian matrix of the following function :

$$f(X) = x_1^2 + 2x_1x_2 + 3x_2^2 + 4x_1x_2x_3 + x_3^3 \ .$$

The Hessian matrix is given by

$$\nabla_X^2 f(X) = \begin{bmatrix} \frac{\partial f}{\partial x_1^2} & \frac{\partial x^2 f}{\partial x_1 x_2} & \frac{\partial f^2}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial x^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3} \end{bmatrix} \in R^{3\times 3}$$

$$H[F(X) = \begin{bmatrix} 2 & 2+4x_3 & 4x_2 \\ 2+4x_3 & 6 & 4x_1 \\ 4x_2 & 4x_1 & 6x_3 \end{bmatrix}_{3\times 3}$$

Note that the Hessian matrix is symmetric.

Check Your Progress 3

1) Given the fact that for any two vectors, **a** and **b**, c $vec\ \boldsymbol{ab}' = \boldsymbol{b} \otimes \boldsymbol{a}$. Show that $vec\ \boldsymbol{ABC} = (\boldsymbol{C}' \otimes \boldsymbol{A})vecB$, whenever the product **ABC** is defined.

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2) Let $x_1, x_2 \ldots x_n$ be an independent and identically distributed (iid) sample from $N(\mu, \sigma^2)$. Let $\theta = (\mu, \sigma^2)$. If log-likelihood function $log\, L\,(\theta) = -\frac{n}{2} ln\, \sigma^2 - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2$, then derive the Fisher information.

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3.12 LET US SUM UP

In this Unit we focussed on multiplication and inversion of matrices. For invertibility, a square matrix must be full rank. Therefore, we discussed about the method of finding the rank of a matrix by the reduced echelon form method. A matrix of large dimension can be partitioned into various blocks, which make calculation easy. Thus, we presented the results of matrix operations of a partitioned matrix. Diagonalization of matrix and eigenvalue of matrix are also discussed. We discussed certain special matrices which are useful in econometrics. In the end, we discussed Kronecker product, vectorization of matrix and matrix differentiation.

3.13 KEYWORDS

Linear Independence of Vectors : A set of n vectors, denoted $\{U_1, U_2, \ldots U_n\}$ is said to be linearly independent if the matrix equation

$$c_1 u_1 + \ldots + c_n u_n = 0$$

has only zero solution $c_1 \ldots = c_n = 0$. If the above equation holds for a set of coefficient that are not all zero, then the n vectors are said to be linearliy independent.

Euclidean Norm or $\boldsymbol{l_2}$ **Norm of a Vector** : $\|\boldsymbol{x_2}\|_2 = \|\boldsymbol{X}\|_E = \left(|\boldsymbol{x_1^2}| + |\boldsymbol{x_2}|^2 \ldots\right)^{\boldsymbol{2}}$**.** A vector with unit Euclidean length is referred to as a normalized vector.

Idempotent Matrix : A matrix $\boldsymbol{A_{n\times n}}$ is idempotent if $\boldsymbol{A^2} = \boldsymbol{AA} = \boldsymbol{A}$. Its eigenvalues are 0 and 1 except for identity matrix but converse is not necessarily true.

Characteristic Polynomial : ${P_{nx}}^n + P_{n-1}x^{n-1} \ldots . P_{1x} + P_0$ is known as the characteristic polynomial of A and $P(x) = det(A - xI) = 0$ is said to be characteristic equation of A.

3.14 SOME USEFUL BOOKS

Abadir. M. Karim and Magnus. R. Jan (2005), *Matrix Algebra*, Cambridge University Press.

Gentle. E. James, 2017, *Matrix Algebra: Theory, Computations and Applications in Statistics*, Second Edition, Springer International Publishing AG.

Magnus, R. Jan, and Neudeckery Heinz, 2019, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, Third Edition, John Wiley & Sons.

Schoot, R. James, 2017, *Matrix Analysis for Statistics*, Third Edition, John Wiley & Sons.

Turking-ton, A Darrell, 2005, *Matrix Calculus and Zero-One Matrices: Statistical and Econometric Applications*, Cambridge University Press.

3.15 ANSWERS/HINT TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) We can rewrite both the equations with endogenous variables C and Y on the left side

$$Y - C = I + G$$

$$-bY + C = a$$

Writing above equations in matrix notation, we obtain

$$\begin{bmatrix} 1 & -1 \\ -b & 1 \end{bmatrix}\begin{bmatrix} Y \\ C \end{bmatrix} = \begin{bmatrix} I + G \\ a \end{bmatrix}$$

This resembles the equation **AX** = **B** where $\boldsymbol{X} = \begin{bmatrix} Y \\ C \end{bmatrix}$

Hence, $\begin{bmatrix} Y \\ C \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -b & 1 \end{bmatrix}^{-1}\begin{bmatrix} I + G \\ a \end{bmatrix}$

$$\Rightarrow X = \begin{bmatrix} Y \\ C \end{bmatrix} = \frac{-1}{1-b}\begin{bmatrix} 1 & 1 \\ b & 1 \end{bmatrix}\begin{bmatrix} I + G \\ a \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} Y \\ C \end{bmatrix} = \begin{bmatrix} \frac{I+G+a}{1-b} \\ \frac{(1+G)b+a}{1-b} \end{bmatrix}.$$

2) $\boldsymbol{A} = \begin{bmatrix} 2 & 4 & 1 \\ 4 & 3 & 7 \\ 2 & 1 & 3 \end{bmatrix}$

First, compute the cofactor of C_{11}. For this, you have to delete row 1 and column 1 of matrix **A**. Next, compute determinant of the residual 2 × 2 matrix.

Thus, $C_{11} = (-1)^{1+1}\begin{vmatrix} 3 & 7 \\ 1 & 3 \end{vmatrix} = 2.$

Similarly, you should compute

$$C_{12} = (-1)^{1+2}\begin{vmatrix} 4 & 7 \\ 2 & 3 \end{vmatrix} = 2$$

$$C_{13} = (-1)^{1+3}\begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix} = -2$$

$$C_{21} = (-1)^{2+1}\begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} = -11$$

$$C_{22} = (-1)^{2+2}\begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} = 4$$

$$C_{23} = (-1)^{2+3}\begin{vmatrix} 2 & 4 \\ 2 & 1 \end{vmatrix} = 6$$

$$C_{31} = (-1)^{3+1}\begin{vmatrix} 4 & 1 \\ 3 & 7 \end{vmatrix} = 25$$

$$C_{32} = (-1)^{3+2}\begin{vmatrix} 2 & 1 \\ 4 & 7 \end{vmatrix} = -10$$

$C_{33} = (-1)^{3+3}\begin{vmatrix}2 & 4\\4 & 3\end{vmatrix} = -10$

The cofactor matrix $= \begin{bmatrix}2 & 2 & -2\\-11 & 4 & 6\\25 & -10 & -10\end{bmatrix}$

$det\, A = a_{21}.C_{21} \quad + \quad a_{22}.C_{22} \quad + \quad a_{23}.C_{23}$

$\quad = (4-11) \quad + \quad (3.4) \quad + \quad 7.6 = 10$

Adjoint matrix = (cofactor matrix)^T

$= \begin{bmatrix}2 & -11 & 25\\2 & 4 & -10\\-2 & 6 & -10\end{bmatrix}$ = adjoint **A**

Therefore, $\boldsymbol{A^{-1}} = \frac{1}{det\,A} adjo\,int\,A = \begin{bmatrix}115 & -\frac{11}{10} & 5/2\\115 & 2/5 & -1\\-1/5 & 3/5 & -1\end{bmatrix}$

3) The matrix for determination of rank is

$$\begin{pmatrix}3 & 0 & 2 & 2\\-6 & 42 & 24 & 54\\21 & -21 & 0 & -15\end{pmatrix}$$

$$\xrightarrow{R_2\to R_2+2R_1}\begin{pmatrix}3 & 0 & 2 & 2\\0 & 42 & 28 & 58\\21 & -21 & 0 & -15\end{pmatrix}$$

$$\xrightarrow{R_3\to R_3-7R_1}\begin{pmatrix}3 & 0 & 2 & 2\\0 & 42 & 28 & 58\\0 & -21 & -14 & -29\end{pmatrix}$$

$$\xrightarrow{R_3\to R_3+\frac{1}{2}R_2}\begin{pmatrix}3 & 0 & 2 & 2\\0 & 42 & 28 & 58\\0 & 0 & 0 & 0\end{pmatrix}$$

Above, we converted the original matrix to row echelon form by applying elementary row operations. The number of non-zero rows is 2; hence the rank of the matrix is 2. Recall that we can interpret rank = no of pivots. Thus, rank = no of pivots = 2.

Check Your Progress 2

1. If **A** has n linearly independent eigenvectors $x_1, x_2 \ldots . x_n$, then $\boldsymbol{T} = (x_1 \ldots x_n)$ is non-singular and satisfies

$$\boldsymbol{AT} = \boldsymbol{A}(x_1, \ldots, x_n) = (\boldsymbol{A}x_1 \ldots . \boldsymbol{A}x_n) = (\lambda_1 x_1 \ldots . \lambda_n x_n)$$

$$= (x_1, x_2 \ldots x_n)\boldsymbol{\Lambda} = \boldsymbol{T\Lambda}$$

where $\boldsymbol{\Lambda} = diag.(\lambda_1, \lambda_2 \ldots . \lambda_n)$ and $\lambda_1, \lambda_2 \ldots \lambda_n$ are eigen values of **A**. Hence $\boldsymbol{T^{-1}AT} = \boldsymbol{\Lambda}$. Conversely, if $\boldsymbol{AT} = \boldsymbol{T\Lambda}$ for some diagonal matrix $\boldsymbol{\Lambda}$, then $\boldsymbol{A}(\boldsymbol{T}e_i) = (\boldsymbol{AT})e_i = (\boldsymbol{T\Lambda})e_i = \boldsymbol{T}(\boldsymbol{\Lambda}e_i) = \boldsymbol{T}(\lambda_i e_i) = \lambda_i(\boldsymbol{T}e_i)$

where e_i denotes the ith unit vector, a vector which has 1 at ith place and zero in all other places.

Hence, each column of **T** is an eigenvector of **A** and since **T** is non-singular, these eigenvectors are linearly independent.

2. (a) It is given that $\boldsymbol{M} = \boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$

We have to show that **M** is symmetric and idempotent. First, the symmetric part. A matrix is symmetric when $\boldsymbol{A} = \boldsymbol{A'}$. By applying this argument $\boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$ is symmetric. Second, the idempotent part. Check whether **MM** = **M**

$$(\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})(\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})$$

$$= \boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} + \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$$

$$= \boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} + \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$$

$$= \boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} = \boldsymbol{M}$$

(b) We have to verify whether $\boldsymbol{MX} = \boldsymbol{0}.$

$$\boldsymbol{MX} = (\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})\boldsymbol{X}$$

$$= \boldsymbol{X} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'X} = \boldsymbol{X} - \boldsymbol{X} = \boldsymbol{0}$$

(c) Using the rule, rank of an idempotent matrix is the trace of that matrix.

rank (**M**) = trace (**M**) = $tr(\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})$

$$= tr(\boldsymbol{I_n}) - tr(\boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})$$

$$= tr(\boldsymbol{I_n}) - tr(\boldsymbol{X'X}(\boldsymbol{X'X})^{-1})$$

$$= n - tr(\boldsymbol{I_k}) = n - k$$

3. We know that $\boldsymbol{X'V^{-1}M} = \boldsymbol{0}$

(a) $\Rightarrow \boldsymbol{X'V^{-1}}(\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}) = \boldsymbol{0}$

$$\Rightarrow \boldsymbol{X'V^{-1}} = \boldsymbol{X'V^{-1}X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$$

$$\Rightarrow (\boldsymbol{X'V^{-1}X})^{-1}\boldsymbol{X'V^{-1}} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'}$$

(b) Post-multiply both sides of equality in the above by $\boldsymbol{VX}(\boldsymbol{X'X})^{-1}$. We get

$$(\boldsymbol{X'V^{-1}X})^{-1}\boldsymbol{X'V^{-1}VX}(\boldsymbol{X'X})^{-1} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'VX}(\boldsymbol{X'X})^{-1}$$

$$\Rightarrow (\boldsymbol{X'V^{-1}X})^{-1} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'VX}(\boldsymbol{X'X})^{-1}$$

Check Your Progress 3

1. Let $\boldsymbol{b} = (b_1 \ldots . b_n)'$Then,

$$vec(\boldsymbol{ab'}) = vec(b_1 a_1 \ldots b_n a) = \begin{pmatrix} b_1 a \\ \vdots \\ b_n a \end{pmatrix} = \boldsymbol{b} \otimes \boldsymbol{a}$$

Let $\boldsymbol{B} = (\boldsymbol{b_1}, \boldsymbol{b_2}, \ldots, \boldsymbol{b_n})$ be a m×n matrix. Let us denote the n column of $\boldsymbol{I_n}$ matrix by $e_1 \ldots e_n$. Then, **B** can be written as $\boldsymbol{B} = \sum_{i=1}^{n} b_i e_i'$

Hence, $vec(\boldsymbol{ABC}) = vec \sum_{i=1}^{n} \boldsymbol{A} b_i e_i' \, \boldsymbol{C} = \sum_{i=1}^{n} vec((\boldsymbol{A}b_i)(C' e_i)')$

$$= \sum_{i=1}^{n} (\boldsymbol{C'} e_i) \otimes (\boldsymbol{A} b_i) = (\boldsymbol{C'} \otimes \boldsymbol{A}) \sum_{i=1}^{n} e_i \otimes b_i$$

$= (\boldsymbol{C}' \otimes \boldsymbol{A}) \sum_{i=1}^{n} vec(b_i e_i') = (\boldsymbol{C}' \otimes A) vec(\boldsymbol{B})$

2. Let us re-state the problem: We have $x_1, x_2, \ldots, x_n$ be iid (independent and identically distributed) sample from normal distribution $N(\mu, \sigma^2)$ where μ =mean, σ^2 =variance

Let the parameter (the unknowns to be estimated) space $\theta = (\mu, \sigma^2)$

The log-likelihood function is

$$log\, L(\theta) = -\frac{n}{2} ln\, \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

The score function is now a gradient vector

$$S(\theta) = \frac{\partial}{\partial \theta} log\, L(\theta) = \begin{pmatrix} \frac{\partial}{\partial \mu} ln\, L(\theta) \\ \frac{\partial}{\partial \sigma^2} ln\, L(\theta) \end{pmatrix} = \begin{pmatrix} \frac{n}{\sigma^2}(\bar{x} - \mu) \\ -\frac{n}{2\sigma^2} + \frac{\sum (x_i - \mu)^2}{2\sigma^4} \end{pmatrix}$$

The observed Fisher information is minus the Hessian of the log-likelihood function. Thus,

$$I(\theta) = -\frac{\partial^2}{\partial \theta \partial \theta'} log\, L(\theta)$$

$$I(\theta) = \begin{pmatrix} \frac{n}{\sigma^2} & \frac{n}{\sigma^4}(\bar{x} - \mu) \\ \frac{n}{\sigma^4}(\bar{x} - \mu) & -\frac{n}{2\sigma^4} + \frac{\sum (x_i - \mu)^2}{(\sigma^2)^3} \end{pmatrix}$$

The expected Fisher information is

$I(\theta) = E_\theta I(\theta)$.

The above is called the information matrix, a 2 × 2 variance matrix and it is a positive semi-definite matrix.

$$I(\theta) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix}$$

$det\, d\,(\theta)$ is always positive because n and σ^2 are always positive.